



Parametric bootstrap methods for bias correction in linear mixed models

Tatsuya Kubokawa^{a,*}, Bui Nagashima^b

^a Faculty of Economics, University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo 113-0033, Japan

^b Graduate School of Economics, University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo 113-0033, Japan

ARTICLE INFO

Article history:

Received 22 April 2011

Available online 16 December 2011

AMS 2010 subject classification:

62F40

62J07

62F12

62F15

Keywords:

Best linear unbiased predictor

Confidence interval

Empirical Bayes procedure

Fay–Herriot model

Second-order correction

Linear mixed model

Maximum likelihood estimator

Mean squared error

Nested error regression model

Parametric bootstrap

Restricted maximum likelihood estimator

Small area estimation

ABSTRACT

The empirical best linear unbiased predictor (EBLUP) in the linear mixed model (LMM) is useful for the small area estimation, and the estimation of the mean squared error (MSE) of EBLUP is important as a measure of uncertainty of EBLUP. To obtain a second-order unbiased estimator of the MSE, the second-order bias correction has been derived based on Taylor series expansions. However, this approach is hard to implement in complicated models with many unknown parameters like variance components, since we need to compute asymptotic bias, variance and covariance for estimators of unknown parameters as well as partial derivatives of some quantities. A similar difficulty occurs in the construction of confidence intervals based on EBLUP with second-order correction and in the derivation of second-order bias correction in the Akaike Information Criterion (AIC) and the conditional AIC. To avoid such difficulty in the derivation of second-order bias correction in these problems, the parametric bootstrap methods are suggested in this paper, and their second-order justifications are established. Finally, performances of the suggested procedures are numerically investigated in comparison with some existing procedures given in the literature.

© 2011 Elsevier Inc. All rights reserved.

1. Introduction

The linear mixed models (LMM) and the model-based estimates including empirical best linear unbiased predictor (EBLUP) or the empirical Bayes estimator (EB) have been recognized useful in small area estimation. The typical models used for the small area estimation are the Fay–Herriot model and the nested error regression model (NERM), and the usefulness of EBLUP is illustrated by Fay and Herriot [15] and Battese et al. [4]. For a good review and account on this topic, see [16,26,24].

When EBLUP is used to estimate a small area mean based on real data, it is important to estimate the mean squared error (MSE) since it can assess how much EBLUP is reliable. Asymptotically unbiased estimators of the MSE with the second-order bias correction have been derived based on the Taylor series expansion by Prasad and Rao [25], Datta and Lahiri [11], Datta et al. [12], Das et al. [8], Kubokawa [20] and others. A drawback of this method is that it is harder to compute the second-order bias, variance and covariance of estimators of more unknown parameters including variance components, and that it is troublesome to derive partial derivatives of some matrices with respect to unknown parameters. To avoid

* Corresponding author.

E-mail addresses: tatsuya@e.u-tokyo.ac.jp (T. Kubokawa), bui-bui.miu-miu@kde.biglobe.ne.jp (B. Nagashima).

this difficulty, Butar and Lahiri [5] proposed the parametric bootstrap method, which is easy to implement, since we do not have to compute the second-order bias, variance and partial derivatives. For some recent results including nonparametric methods, see [22,17,7].

In the construction of a confidence interval based on EBLUP with the second-order accuracy, we are faced with a similar problem. Basu et al. [3], Datta et al. [9], Kubokawa [19] derived such confidence intervals using the Taylor series expansion. To avoid the difficulty in derivation of second-order moments, Chatterjee et al. [6], Hall and Maiti [18] proposed the confidence intervals using the parametric bootstrap method.

A similar difficulty occurs in evaluating the bias terms of *AIC* and conditional *AIC*. The Akaike Information Criterion (*AIC*) originated from [1,2] is recognized very useful for selecting models in general situations, and it is also useful for selecting variables in LMM. When unknown parameters in the model are estimated by the maximum likelihood estimator, the penalty term, which is a kind of bias, is known to be $2 \times p$ for dimension p of unknown parameters. When the unknown parameters in LMM are estimated by other estimators, however, Kubokawa [21] showed that the penalty term includes partial derivatives of the estimator and the covariance matrix. Concerning the conditional *AIC*, on the other hand, Liang et al. [23] Vaida and Blanchard [29] proposed the conditional *AIC* in LMM, but their derivations were limited to the cases that the parameters in LMM are partly known. Recently, Kubokawa [21] derived the second-order bias correction for the conditional *AIC*, but it is harder to compute in more complicated models.

In this paper, we treat the problems mentioned above, and provide useful procedures based on the parametric bootstrap methods to avoid the computational difficulties. In Section 2, we suggest the MSE estimator, the confidence interval, *AIC* and the conditional *AIC* using the parametric bootstrap. Concerning the MSE estimation, Butar and Lahiri [5] estimated the third term of the MSE, denoted by g_3 , based on the parametric bootstrap, while in this paper, we consider to estimate the second-order approximation of g_3 using the parametric bootstrap method. A similar approach applies to the confidence interval, and we estimate the second-order correction term based on the parametric bootstrap method. This is different from the parametric bootstrap procedure suggested by Chatterjee et al. [6] who obtained two end-points of a confidence interval based on a distribution generated by the parametric bootstrap sampling. Simulation and empirical studies are given in Section 3. The proofs for the second-order justifications of the proposed procedures are given in the [Appendix](#).

2. MSE estimation, confidence interval and *AIC* based on the parametric bootstrap method

2.1. Linear mixed model and the parametric bootstrap method

Consider the following general linear mixed model.

[1] *Model 1.* An $N \times 1$ observation vector $\mathbf{y}$ of the response variable has the model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{v} + \boldsymbol{\epsilon}, \quad (1)$$

where $\mathbf{X}$ and $\mathbf{Z}$ are $N \times p$ and $N \times M$ matrices, respectively, of the explanatory variables, $\boldsymbol{\beta}$ is a $p \times 1$ unknown vector of the regression coefficients, $\mathbf{v}$ is an $M \times 1$ vector of the random effects, and $\boldsymbol{\epsilon}$ is an $N \times 1$ vector of the random errors. Here, $\mathbf{v}$ and $\boldsymbol{\epsilon}$ are mutually independently distributed as $\mathbf{v} \sim \mathcal{N}_M(\mathbf{0}, \mathbf{G}(\boldsymbol{\theta}))$ and $\boldsymbol{\epsilon} \sim \mathcal{N}_N(\mathbf{0}, \mathbf{R}(\boldsymbol{\theta}))$, where $\boldsymbol{\theta} = (\theta_1, \dots, \theta_q)'$ is a q -dimensional vector of unknown parameters, and $\mathbf{G} = \mathbf{G}(\boldsymbol{\theta})$ and $\mathbf{R} = \mathbf{R}(\boldsymbol{\theta})$ are positive definite matrices. Then, $\mathbf{y}$ has a marginal distribution $\mathcal{N}_N(\mathbf{X}\boldsymbol{\beta}, \boldsymbol{\Sigma}(\boldsymbol{\theta}))$ for

$$\boldsymbol{\Sigma} = \boldsymbol{\Sigma}(\boldsymbol{\theta}) = \mathbf{R}(\boldsymbol{\theta}) + \mathbf{Z}\mathbf{G}(\boldsymbol{\theta})\mathbf{Z}'.$$

Throughout the paper, we often drop $(\boldsymbol{\theta})$ in $\mathbf{G}(\boldsymbol{\theta})$, $\mathbf{R}(\boldsymbol{\theta})$, $\boldsymbol{\Sigma}(\boldsymbol{\theta})$ and others for notational convenience.

For simplicity, it is here assumed that $\mathbf{X}$ is of full rank. However, the results given in this paper can be extended to the case that $\mathbf{X}$ is not of full rank by modifying $\mathbf{X}$ and $\boldsymbol{\beta}$ as follows. Let r be a rank of $\mathbf{X}$, and suppose that $r < p$. Then, $\mathbf{X}$ can be written as

$$\mathbf{X} = \mathbf{P} \begin{pmatrix} \mathbf{E}_r & \mathbf{0} \\ \mathbf{0}' & \mathbf{0} \end{pmatrix} \mathbf{Q},$$

where $\mathbf{P}$ and $\mathbf{Q}$ are, respectively, $N \times N$ and $p \times p$ orthogonal matrices and $\mathbf{E}_r = \text{diag}(\lambda_1, \dots, \lambda_r)$ for positive constants λ_i 's. Let $\mathbf{P} = (\mathbf{P}_1, \mathbf{P}_2)$ and $\mathbf{Q}' = (\mathbf{Q}'_1, \mathbf{Q}'_2)$ for $N \times r$ matrix $\mathbf{P}_1$ and $r \times p$ matrix $\mathbf{Q}_1$. Thus, $\mathbf{X}\boldsymbol{\beta}$ can be rewritten as

$$\mathbf{X}\boldsymbol{\beta} = \mathbf{P}_1\mathbf{E}_r\mathbf{Q}_1\boldsymbol{\beta}.$$

Hence, all the results given in this paper still hold if we replace $(\mathbf{X}, \boldsymbol{\beta}, p)$ with $(\mathbf{P}_1\mathbf{E}_r, \mathbf{Q}_1\boldsymbol{\beta}, r)$.

The unknown parameters in Model 1 are $\boldsymbol{\beta}$ and $\boldsymbol{\theta}$. When $\boldsymbol{\theta}$ is known, the regression coefficients vector $\boldsymbol{\beta}$ is estimated by the generalized least squares estimator given by

$$\widehat{\boldsymbol{\beta}}(\boldsymbol{\theta}) = (\mathbf{X}'\boldsymbol{\Sigma}(\boldsymbol{\theta})^{-1}\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Sigma}(\boldsymbol{\theta})^{-1}\mathbf{y}.$$

The parameter $\boldsymbol{\theta}$ consists of variance components and others, and it is estimated by consistent estimator $\widehat{\boldsymbol{\theta}}$ based on $\mathbf{y}$ which can be constructed by maximum likelihood, restricted maximum likelihood and other methods. Then, $\boldsymbol{\beta}$ is estimated by $\widehat{\boldsymbol{\beta}} = \widehat{\boldsymbol{\beta}}(\widehat{\boldsymbol{\theta}})$.

We next consider the following conditional linear mixed model given $\mathbf{y}$.

(2) *Model 2.* An $N \times 1$ random vector $\mathbf{y}^*$ has the model

$$\mathbf{y}^* = \mathbf{X}\hat{\boldsymbol{\beta}}(\hat{\boldsymbol{\theta}}) + \mathbf{Z}\mathbf{v}^* + \boldsymbol{\epsilon}^*, \quad (2)$$

where $\mathbf{X}$ and $\mathbf{Z}$ are the same matrices as given in (1), and given $\mathbf{y}$, the random vectors $\mathbf{v}^*$ and $\boldsymbol{\epsilon}^*$ are conditionally mutually independently distributed as $\mathbf{v}^*|\mathbf{y} \sim \mathcal{N}_M(\mathbf{0}, \mathbf{G}(\hat{\boldsymbol{\theta}}))$ and $\boldsymbol{\epsilon}^*|\mathbf{y} \sim \mathcal{N}_N(\mathbf{0}, \mathbf{R}(\hat{\boldsymbol{\theta}}))$.

Before stating the main results, we briefly explain the intuitive idea of the parametric bootstrap method based on Model 2. Let $g(\boldsymbol{\theta})$ be a differentiable function with $g(\boldsymbol{\theta}) = O(1)$. Although $g(\hat{\boldsymbol{\theta}})$ is an asymptotically unbiased estimator of $g(\boldsymbol{\theta})$, in general, there exists a second-order bias. Then, we need to approximate the expectation $E[g(\hat{\boldsymbol{\theta}})]$ up to $O(N^{-1})$. It is supposed that the approximation is given by

$$E[g(\hat{\boldsymbol{\theta}})] = g(\boldsymbol{\theta}) + b_g(\boldsymbol{\theta}) + O(N^{-3/2}), \quad (3)$$

where $b_g(\boldsymbol{\theta})$ is a continuously differentiable function with $O(N^{-1})$. Then,

$$\begin{aligned} E[g(\hat{\boldsymbol{\theta}}) - b_g(\hat{\boldsymbol{\theta}})] &= E[g(\hat{\boldsymbol{\theta}})] - E[b_g(\hat{\boldsymbol{\theta}})] \\ &= \{g(\boldsymbol{\theta}) + b_g(\boldsymbol{\theta})\} - b_g(\boldsymbol{\theta}) + O(N^{-3/2}) = g(\boldsymbol{\theta}) + O(N^{-3/2}). \end{aligned} \quad (4)$$

Using Model 2, from (3), we can see that

$$E_*[g(\hat{\boldsymbol{\theta}}^*)|\mathbf{y}] = g(\hat{\boldsymbol{\theta}}) + b_g(\hat{\boldsymbol{\theta}}) + O_p(N^{-3/2}),$$

where $E_*[\cdot|\mathbf{y}]$ is the conditional expectation with respect to Model 2 given $\mathbf{y}$, and the calculation of $\hat{\boldsymbol{\theta}}^*$ is the same as that of $\hat{\boldsymbol{\theta}}$ except that $\hat{\boldsymbol{\theta}}^*$ is calculated based on $\mathbf{y}^*$ instead of $\mathbf{y}$. Hence from (4), it is seen that

$$\begin{aligned} E[2g(\hat{\boldsymbol{\theta}}) - E_*[g(\hat{\boldsymbol{\theta}}^*)|\mathbf{y}]] &= E[g(\hat{\boldsymbol{\theta}}) - E_*[g(\hat{\boldsymbol{\theta}}^*) - g(\hat{\boldsymbol{\theta}})|\mathbf{y}]] \\ &= E[g(\hat{\boldsymbol{\theta}}) - b_g(\hat{\boldsymbol{\theta}})] + O(N^{-3/2}) \\ &= g(\boldsymbol{\theta}) + O(N^{-3/2}). \end{aligned} \quad (5)$$

Thus, we obtain the second-order unbiased estimator $2g(\hat{\boldsymbol{\theta}}) - E_*[g(\hat{\boldsymbol{\theta}}^*)|\mathbf{y}]$.

Since, in general, the function $b_g(\boldsymbol{\theta})$ can be derived by the Taylor series expansion, it includes partial derivatives with respect to θ_i , $i = 1, \dots, q$, and moments of estimator $\hat{\boldsymbol{\theta}}$. It is clear that the second-order unbiased estimator $2g(\hat{\boldsymbol{\theta}}) - E_*[g(\hat{\boldsymbol{\theta}}^*)|\mathbf{y}]$ is free from differentiations or moments of $\hat{\boldsymbol{\theta}}$. This idea was used by Butar and Lahiri [5], Chatterjee et al. [6] and others in the framework of small area estimation, and we heavily employ the parametric bootstrap method in this paper.

2.2. Estimation of MSE of EBLUP

Based on the parametric bootstrap method, we first derive an estimator of MSE of EBLUP for the general scalar quantity

$$\mu = \mathbf{a}'\boldsymbol{\beta} + \mathbf{b}'\mathbf{v},$$

where $\mathbf{a}$ and $\mathbf{b}$ be $p \times 1$ and $M \times 1$ vectors of fixed constants. It is noted that the marginal and the conditional distributions of $\mathbf{y}$ given $\mathbf{v}$ are, respectively,

$$\mathbf{y} \sim \mathcal{N}_N(\mathbf{X}\boldsymbol{\beta}, \boldsymbol{\Sigma}(\boldsymbol{\theta})),$$

$$\mathbf{y}|\mathbf{v} \sim \mathcal{N}_N(\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{v}, \mathbf{R}(\boldsymbol{\theta})).$$

Let $\boldsymbol{\mu}_v = \mathbf{G}\mathbf{Z}'\boldsymbol{\Sigma}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$ and $\boldsymbol{\Sigma}_v = \mathbf{G} - \mathbf{G}\mathbf{Z}'\boldsymbol{\Sigma}^{-1}\mathbf{Z}\mathbf{G}$. Since the conditional distribution of $\mathbf{v}$ given $\mathbf{y}$ is

$$\mathbf{v}|\mathbf{y} \sim \mathcal{N}_M(\boldsymbol{\mu}_v, \boldsymbol{\Sigma}_v), \quad (6)$$

the conditional expectation $E[\mu|\mathbf{y}]$ is written as

$$\hat{\mu}^B(\boldsymbol{\beta}, \boldsymbol{\theta}) = \mathbf{a}'\boldsymbol{\beta} + \mathbf{s}(\boldsymbol{\theta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}), \quad (7)$$

where $\mathbf{s}(\boldsymbol{\theta}) = \boldsymbol{\Sigma}(\boldsymbol{\theta})^{-1}\mathbf{Z}\mathbf{G}(\boldsymbol{\theta})\mathbf{b}$. This can be interpreted as the Bayes estimator of μ in the Bayesian context. The generalized least squares estimator $\hat{\boldsymbol{\beta}}(\boldsymbol{\theta})$ is substituted into $\hat{\mu}^B(\boldsymbol{\beta}, \boldsymbol{\theta})$ to get the estimator

$$\hat{\mu}^{EB}(\boldsymbol{\theta}) = \hat{\mu}^B(\hat{\boldsymbol{\beta}}(\boldsymbol{\theta}), \boldsymbol{\theta}) = \mathbf{a}'\hat{\boldsymbol{\beta}}(\boldsymbol{\theta}) + \mathbf{s}(\boldsymbol{\theta})'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}(\boldsymbol{\theta})), \quad (8)$$

which is the best linear unbiased predictor (BLUP) of μ . When an estimator $\hat{\boldsymbol{\theta}}$ is available for $\boldsymbol{\theta}$, we can estimate μ by the empirical (or estimated) best linear unbiased predictor (EBLUP) $\hat{\mu}^{EB}(\hat{\boldsymbol{\theta}})$, which is also called an empirical Bayes estimator in the Bayesian context.

The MSE function of EBLUP $\hat{\mu}^{EB}(\hat{\theta})$ is $MSE(\theta, \hat{\mu}^{EB}(\hat{\theta})) = E[\{\hat{\mu}^{EB}(\hat{\theta}) - \mu\}^2]$, which can be decomposed as $MSE(\theta, \hat{\mu}^{EB}(\hat{\theta})) = g_1(\theta) + g_2(\theta) + g_3(\theta)$ as shown in [25,11], where $g_1(\theta) = E[\{\hat{\mu}^B(\beta, \theta) - \mu\}^2]$, $g_2(\theta) = E[\{\hat{\mu}^{EB}(\theta) - \hat{\mu}^B(\beta, \theta)\}^2]$ and $g_3(\theta) = E[\{\hat{\mu}^{EB}(\hat{\theta}) - \hat{\mu}^{EB}(\theta)\}^2]$. The terms $g_1(\theta)$ and $g_2(\theta)$ can be rewritten as

$$\begin{aligned} g_1(\theta) &= \mathbf{b}'(\mathbf{G}(\theta)^{-1} + \mathbf{Z}'\mathbf{R}(\theta)^{-1}\mathbf{Z})^{-1}\mathbf{b}, \\ g_2(\theta) &= (\mathbf{a} - \mathbf{X}'\mathbf{s}(\theta))'(\mathbf{X}'\mathbf{\Sigma}(\theta)^{-1}\mathbf{X})^{-1}(\mathbf{a} - \mathbf{X}'\mathbf{s}(\theta)). \end{aligned}$$

Using the argument as in (5), we can estimate $g_1(\theta) + g_2(\theta)$ by

$$2\{g_1(\hat{\theta}) + g_2(\hat{\theta})\} - E_*[g_1(\hat{\theta}^*) + g_2(\hat{\theta}^*)|\mathbf{y}].$$

For $g_3(\theta)$, in this paper we use the estimator given by

$$\bar{g}_3^*(\hat{\theta}) = E_*[\{\mathbf{s}(\hat{\theta}^*) - \mathbf{s}(\hat{\theta})\}'\mathbf{\Sigma}(\hat{\theta})\{\mathbf{s}(\hat{\theta}^*) - \mathbf{s}(\hat{\theta})\}|\mathbf{y}]. \quad (9)$$

Thus, we get the estimator

$$mse^*(\hat{\theta}, \hat{\mu}^{EB}(\hat{\theta})) = 2\{g_1(\hat{\theta}) + g_2(\hat{\theta})\} - E_*[g_1(\hat{\theta}^*) + g_2(\hat{\theta}^*)|\mathbf{y}] + \bar{g}_3^*(\hat{\theta}). \quad (10)$$

In the [Appendix](#), we shall show that $E[mse^*(\hat{\theta}, \hat{\mu}^{EB}(\hat{\theta}))] = MSE(\theta, \hat{\mu}^{EB}(\hat{\theta})) + O(N^{-3/2})$ under some conditions.

2.3. Confidence interval based on EBLUP

We next construct a confidence interval of $\mu = \mathbf{a}'\beta + \mathbf{b}'\mathbf{v}$ based on EBLUP which satisfies the nominal confidence level with the second-order accuracy. Since $mse^*(\hat{\theta}, \hat{\mu}^{EB}(\hat{\theta}))$ is an asymptotically unbiased estimator of the MSE of EBLUP, it is reasonable to consider the confidence interval of the form

$$I^{EB}(\hat{\theta}): \hat{\mu}^{EB}(\hat{\theta}) \pm z_{\alpha/2} \sqrt{\max\{mse^*(\hat{\theta}, \hat{\mu}^{EB}(\hat{\theta})), 0\}}, \quad (11)$$

where $z_{\alpha/2}$ is the $100 \times \alpha/2\%$ upper quantile of the standard normal distribution. However, the coverage probability $P[\mu \in I^{EB}(\hat{\theta})]$ cannot be guaranteed to be greater than or equal to the nominal confidence coefficient $1 - \alpha$. To address the problem, we consider the correction function given by

$$h_1^*(\hat{\theta}) = \frac{1 + z_{\alpha/2}^2}{8g_1(\hat{\theta})^2} E_*[\{g_1(\hat{\theta}^*) - g_1(\hat{\theta})\}^2|\mathbf{y}], \quad (12)$$

which is an asymptotically unbiased estimator of $h_1(\theta)$ given in (A.12). Then, the corrected confidence interval is provided by

$$I_1^{CEB*}(\hat{\theta}): \hat{\mu}^{EB}(\hat{\theta}) \pm z_{\alpha/2} \left[1 + h_1^*(\hat{\theta})\right] \sqrt{\max\{mse^*(\hat{\theta}, \hat{\mu}^{EB}(\hat{\theta})), 0\}}. \quad (13)$$

A drawback of $I_1^{CEB*}(\hat{\theta})$ is that it cannot give an interval when $mse^*(\hat{\theta}, \hat{\mu}^{EB}(\hat{\theta}))$ takes a negative value. A simulation experiment given in Section 3.2 shows that such a shortcoming occurs in an extreme case. Thus, we suggest the alternative corrected confidence interval

$$I_2^{CEB*}(\hat{\theta}): \hat{\mu}^{EB}(\hat{\theta}) \pm z_{\alpha/2} \left[1 + h_2^*(\hat{\theta})\right] \sqrt{g_1(\hat{\theta}) + g_2(\hat{\theta})}, \quad (14)$$

where

$$h_2^*(\hat{\theta}) = h_1^*(\hat{\theta}) + \frac{g_1(\hat{\theta}) - E_*[g_1(\hat{\theta}^*)|\mathbf{y}] + \bar{g}_3^*(\hat{\theta})}{2g_1(\hat{\theta})}. \quad (15)$$

In the [Appendix](#), it can be shown that $P[\mu \in I_i^{CEB*}(\hat{\theta})] = 1 - \alpha + O(N^{-3/2})$ for $i = 1, 2$.

2.4. AIC and conditional AIC

We here provide the Akaike Information Criterion (AIC) and conditional AIC based on the parametric bootstrap method. [1] AIC_1^* and AIC_2^* . Let us define the Akaike Information (AI) by

$$AI(\theta) = -2 \int \int \{\log f_m(\tilde{\mathbf{y}}|\hat{\beta}(\mathbf{y}), \hat{\theta}(\mathbf{y}))\} f_m(\tilde{\mathbf{y}}|\beta, \theta) f_m(\mathbf{y}|\beta, \theta) d\tilde{\mathbf{y}} d\mathbf{y}, \quad (16)$$

where $\hat{\beta}(\mathbf{y}) = \hat{\beta}(\hat{\theta})$ and $\hat{\theta}(\mathbf{y})$ are estimators based on $\mathbf{y}$, and $f_m(\mathbf{y}|\beta, \theta)$ is a marginal density function of $\mathbf{y}$ given by

$$-2 \log f_m(\mathbf{y}|\beta, \theta) = N \log(2\pi) + \log |\mathbf{\Sigma}(\theta)| + (\mathbf{y} - \mathbf{X}\beta)' \mathbf{\Sigma}(\theta)^{-1} (\mathbf{y} - \mathbf{X}\beta).$$

Akaike's AIC can be derived as an asymptotically unbiased estimator of $AI(\boldsymbol{\theta})$, namely, $E[AIC] = AI(\boldsymbol{\theta}) + o(1)$ as $N \rightarrow \infty$. When AIC is an exact unbiased estimator of $AI(\boldsymbol{\theta})$, it is called the exact AIC , which was suggested by Sugiura [28], but in general, it is difficult to get in LMM.

Define $\Delta_1^*(\mathbf{y})$ and $\Delta_2^*(\mathbf{y})$ by

$$\begin{aligned}\Delta_1^*(\mathbf{y}) &= -2E_*[\mathbf{u}'\mathbf{P}(\hat{\boldsymbol{\theta}}^*)\mathbf{u}^*|\mathbf{y}], \\ \Delta_2^*(\mathbf{y}) &= E_*[\mathbf{u}'\hat{\boldsymbol{\Sigma}}^{*-1}\mathbf{u}^* - \text{tr}[\hat{\boldsymbol{\Sigma}}\hat{\boldsymbol{\Sigma}}^{*-1}|\mathbf{y}],\end{aligned}\quad (17)$$

where $\mathbf{u}^* = \mathbf{y}^* - \mathbf{X}\hat{\boldsymbol{\beta}}(\hat{\boldsymbol{\theta}})$, $\hat{\boldsymbol{\Sigma}} = \boldsymbol{\Sigma}(\hat{\boldsymbol{\theta}})$, $\hat{\boldsymbol{\Sigma}}^* = \boldsymbol{\Sigma}(\hat{\boldsymbol{\theta}}^*)$ and

$$\mathbf{P}(\boldsymbol{\theta}) = \boldsymbol{\Sigma}(\boldsymbol{\theta})^{-1}\mathbf{X}(\mathbf{X}'\boldsymbol{\Sigma}(\boldsymbol{\theta})^{-1}\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Sigma}(\boldsymbol{\theta})^{-1}.$$

Then, we suggest two kinds of AIC given by

$$\begin{aligned}AIC_1^* &= -2\log f_m(\mathbf{y}|\hat{\boldsymbol{\beta}}(\hat{\boldsymbol{\theta}}), \hat{\boldsymbol{\theta}}) + 2p - \Delta_2^*(\mathbf{y}), \\ AIC_2^* &= -2\log f_m(\mathbf{y}|\hat{\boldsymbol{\beta}}(\hat{\boldsymbol{\theta}}), \hat{\boldsymbol{\theta}}) - \Delta_1^*(\mathbf{y}) - \Delta_2^*(\mathbf{y}).\end{aligned}\quad (18)$$

In the [Appendix](#), it can be shown that $E[AIC_i^*] = AI(\boldsymbol{\theta}) + O(N^{-1/2})$ for $i = 1, 2$.

[2] $cAIC^*$. AIC is derived from the marginal (or unconditional) distribution of $\mathbf{y}$, and it measures the prediction error of the predictor based on the marginal distribution. Since the marginal distribution is integrated out with respect to random effects, EBLUP does not appear in AIC . Vaida and Blanchard [29] proposed the conditional AIC as a criterion incorporating EBLUP.

The conditional AIC is derived as an (asymptotically) unbiased estimator of the conditional Akaike information (cAI) defined by

$$cAI(\boldsymbol{\theta}) = -2 \iint \log\{f(\tilde{\mathbf{y}}|\tilde{\mathbf{v}}(\mathbf{y}), \hat{\boldsymbol{\beta}}(\mathbf{y}), \hat{\boldsymbol{\theta}}(\mathbf{y}))\}f(\tilde{\mathbf{y}}|\mathbf{v}, \boldsymbol{\beta}, \boldsymbol{\theta})f(\mathbf{y}|\mathbf{v}, \boldsymbol{\beta}, \boldsymbol{\theta})f(\mathbf{v}|\boldsymbol{\theta})d\tilde{\mathbf{y}}d\mathbf{y}d\mathbf{v},$$

where $\hat{\boldsymbol{\beta}}(\mathbf{y}) = \hat{\boldsymbol{\beta}}(\hat{\boldsymbol{\theta}})$ and $\hat{\mathbf{v}}(\mathbf{y}) = \hat{\mathbf{v}}(\hat{\boldsymbol{\theta}}) = \mathbf{G}(\boldsymbol{\theta})\mathbf{Z}'\boldsymbol{\Sigma}(\boldsymbol{\theta})^{-1}\{\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}(\boldsymbol{\theta})\}$ are estimators based on $\mathbf{y}$, and $f(\mathbf{y}|\mathbf{v}, \boldsymbol{\beta}, \boldsymbol{\theta})$ and $f(\mathbf{v}|\boldsymbol{\theta})$, respectively, are a conditional density function of $\mathbf{y}$ given $\mathbf{v}$ and a marginal density function of $\mathbf{v}$. Note that

$$-2\log f(\mathbf{y}|\mathbf{v}, \boldsymbol{\beta}, \boldsymbol{\theta}) = N\log(2\pi) + \log|\mathbf{R}(\boldsymbol{\theta})| + (\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{v})'\mathbf{R}(\boldsymbol{\theta})^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{v}).\quad (19)$$

Define $\Delta_{c1}^*(\mathbf{y})$ and $\Delta_{c2}^*(\mathbf{y})$ by

$$\begin{aligned}\Delta_{c1}^*(\mathbf{y}) &= -2\left\{E_*[\mathbf{u}'\hat{\boldsymbol{\Sigma}}^{-1}\hat{\mathbf{R}}\mathbf{P}(\hat{\boldsymbol{\theta}}^*)\mathbf{u}^* + \text{tr}[\hat{\mathbf{R}}^*\hat{\boldsymbol{\Sigma}}^{*-1}|\mathbf{y}] + N - 2\text{tr}[\hat{\mathbf{R}}\hat{\boldsymbol{\Sigma}}^{-1}]\right\}, \\ \Delta_{c2}^*(\mathbf{y}) &= E_*[\mathbf{u}'(2\hat{\boldsymbol{\Sigma}}^{-1}\hat{\mathbf{R}}\hat{\boldsymbol{\Sigma}}^{*-1} - \hat{\boldsymbol{\Sigma}}^{-1}\hat{\mathbf{R}}\hat{\mathbf{R}}^{*-1}\hat{\mathbf{R}}\hat{\boldsymbol{\Sigma}}^{-1})\mathbf{u}^* - \text{tr}[\hat{\mathbf{R}}^{*-1}(2\hat{\mathbf{R}} - \hat{\mathbf{R}}\hat{\boldsymbol{\Sigma}}^{-1}\hat{\mathbf{R}})]|\mathbf{y}] + 2N - 2\text{tr}[\hat{\boldsymbol{\Sigma}}^{-1}\hat{\mathbf{R}}],\end{aligned}\quad (20)$$

for $\hat{\mathbf{R}} = \mathbf{R}(\hat{\boldsymbol{\theta}})$ and $\hat{\mathbf{R}}^* = \mathbf{R}(\hat{\boldsymbol{\theta}}^*)$. Then, we propose the conditional AIC given by

$$cAIC^* = -2\log f(\mathbf{y}|\hat{\mathbf{v}}(\hat{\boldsymbol{\theta}}), \hat{\boldsymbol{\beta}}(\hat{\boldsymbol{\theta}}), \hat{\boldsymbol{\theta}}) - \Delta_{c1}^*(\mathbf{y}) - \Delta_{c2}^*(\mathbf{y}).\quad (21)$$

In the [Appendix](#), it can be shown that $E[cAIC^*] = cAI(\boldsymbol{\theta}) + O(N^{-1/2})$.

3. Simulation and empirical studies

In this section, we investigate performances of the proposed procedures through simulation and empirical studies.

3.1. Fay–Herriot model and procedures used for comparison

In the present and next subsections, we treat the Fay–Herriot model and compare the proposed procedures with ones given in the literature by simulation. The basic area level model proposed by Fay and Herriot [15] is described by

$$y_a = \mathbf{x}_a'\boldsymbol{\beta} + v_a + \varepsilon_a, \quad a = 1, \dots, k,\quad (22)$$

where k is the number of small areas, $\mathbf{x}_a$ is a $p \times 1$ vector of explanatory variables, $\boldsymbol{\beta}$ is a $p \times 1$ unknown common vector of regression coefficients, and v_a 's and ε_a 's are mutually independently distributed random errors such that $v_a \sim \mathcal{N}(0, \theta)$ and $\varepsilon_a \sim \mathcal{N}(0, d_a)$ for known d_a 's. Let $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_k)'$, $\mathbf{y} = (y_1, \dots, y_k)'$, and let $\mathbf{v}$ and $\boldsymbol{\epsilon}$ be similarly defined. Then, the model is expressed in vector notations as $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{v} + \boldsymbol{\epsilon}$ and $\mathbf{y} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}(\boldsymbol{\theta}) = \boldsymbol{\theta}\mathbf{I}_k + \mathbf{D}$ for $\mathbf{D} = \text{diag}(d_1, \dots, d_k)$. In this model, $\mathbf{R} = \mathbf{D}$ and $\mathbf{G} = \boldsymbol{\theta}\mathbf{I}_k$.

For an estimator $\hat{\boldsymbol{\theta}}$ of $\boldsymbol{\theta}$, Model 2 in (2) is described as

$$y_a^* = \mathbf{x}_a'\hat{\boldsymbol{\beta}}(\hat{\boldsymbol{\theta}}) + v_a^* + \varepsilon_a^*, \quad a = 1, \dots, k,\quad (23)$$

where v_a^* 's and ε_a^* 's are mutually independently distributed random errors such that $v_a^*|\mathbf{y} \sim \mathcal{N}(0, \hat{\theta})$ and $\varepsilon_a^* \sim \mathcal{N}(0, d_a)$ for known d_a 's. The estimators $\hat{\theta}^*$ and $\hat{\boldsymbol{\beta}}^*(\hat{\theta}^*)$ can be obtained from $\mathbf{y}_a^*$, $a = 1, \dots, k$, by using the same techniques used to obtain $\hat{\theta}$ and $\hat{\boldsymbol{\beta}}(\hat{\theta})$.

[1] *Estimation of MSE of EBLUP*. It is supposed that we want to predict $\mu_s = \mathbf{x}_s' \boldsymbol{\beta} + v_s$ for some index s among $1, \dots, k$, namely, the vectors $\mathbf{a}$ and $\mathbf{b}$ used in Section 2.2 correspond to $\mathbf{a} = \mathbf{x}_s$ and $\mathbf{b} = \mathbf{j}_s$ where $\mathbf{j}_s = (0, \dots, 0, 1, 0, \dots, 0)'$, namely, the s -th element is one, and the other elements are zero. EBLUP of μ_s is written as

$$\hat{\mu}_s^{EB} = \hat{\mu}_s^{EB}(\mathbf{y}_s; \hat{\boldsymbol{\beta}}(\hat{\theta}), \hat{\theta}) = \mathbf{x}_s' \hat{\boldsymbol{\beta}}(\hat{\theta}) + \{1 - \gamma_s(\hat{\theta})\}(\mathbf{y}_s - \mathbf{x}_s' \hat{\boldsymbol{\beta}}(\hat{\theta})),$$

for $\gamma_s(\theta) = d_s/(\theta + d_s)$, and the functions $\mathbf{s}(\theta)$, $g_1(\theta)$ and $g_2(\theta)$ are expressed as $\mathbf{s}(\theta) = \{1 - \gamma_s(\theta)\}\mathbf{j}_s$, $g_1(\theta) = d_s\{1 - \gamma_s(\theta)\}$ and $g_2(\theta) = \gamma_s(\theta)^2 \mathbf{x}_s' (\mathbf{X}' \boldsymbol{\Sigma}^{-1} \mathbf{X})^{-1} \mathbf{x}_s$.

Concerning the estimation of the MSE of EBLUP $\hat{\mu}_s^{EB}(\mathbf{y}_s; \hat{\boldsymbol{\beta}}(\hat{\theta}), \hat{\theta})$, we here handle the following four estimators: One is the MSE estimator based on the parametric bootstrap method given by

$$mse^*(\hat{\theta}, \hat{\mu}_s^{EB}) = 2\{g_1(\hat{\theta}) + g_2(\hat{\theta})\} - E_*[g_1(\hat{\theta}^*) + g_2(\hat{\theta}^*)|\mathbf{y}] + \bar{g}_3^*(\hat{\theta}), \quad (24)$$

which is from (10), where $\bar{g}_3^*(\hat{\theta}) = E_*[\{\gamma_s(\hat{\theta}^*) - \gamma_s(\hat{\theta})\}^2(\hat{\theta} + d_s)|\mathbf{y}]$. Butar and Lahiri [5] suggested another MSE estimator based on the parametric bootstrap method given by

$$mse^{BL}(\hat{\theta}, \hat{\mu}_s^{EB}) = 2\{g_1(\hat{\theta}) + g_2(\hat{\theta})\} - E_*[g_1(\hat{\theta}^*) + g_2(\hat{\theta}^*)|\mathbf{y}] + E_*[\{\hat{\mu}_s^{EB}(\mathbf{y}_s; \hat{\boldsymbol{\beta}}(\hat{\theta}^*), \hat{\theta}^*) - \hat{\mu}_s^{EB}(\mathbf{y}_s; \hat{\boldsymbol{\beta}}(\hat{\theta}), \hat{\theta})\}^2|\mathbf{y}], \quad (25)$$

Prasad and Rao [25] and Datta and Lahiri [11] suggested the MSE estimator based on the Taylor series expansion given by

$$mse(\hat{\theta}, \hat{\mu}_s^{EB}) = g_1(\hat{\theta}) + g_2(\hat{\theta}) + 2\{\gamma_s(\hat{\theta})^3/d_s\}V(\hat{\theta}) - \{\gamma_s(\hat{\theta})\}^2 B(\hat{\theta}), \quad (26)$$

for $B(\theta) = E[\hat{\theta}^{\dagger\dagger}]$ and $V(\theta) = \text{Var}(\hat{\theta}^\dagger)$. This can be also derived from (A.4). In this model, an exact unbiased estimator of the MSE can be derived by using the Stein identity and [10] provided the unbiased estimator given by

$$mse^E(\hat{\theta}, \hat{\mu}_s^{EB}) = d_s - 2d_s \frac{\partial}{\partial \gamma_s} \left[\gamma_s(\hat{\theta}) \{\mathbf{y}_s - \mathbf{x}_s' \hat{\boldsymbol{\beta}}(\hat{\theta})\} \right] + \{\gamma_s(\hat{\theta})\}^2 \{\mathbf{y}_s - \mathbf{x}_s' \hat{\boldsymbol{\beta}}(\hat{\theta})\}^2. \quad (27)$$

[2] *Corrected confidence interval*. We here treat the following four confidence intervals based on EBLUP: The confidence intervals (13) and (14) with the correction terms using the parametric bootstrap method written by

$$I_1^{CEB*}(\hat{\theta}): \hat{\mu}_s^{EB}(\hat{\theta}) \pm z_{\alpha/2} \left[1 + h_1^*(\hat{\theta}) \right] \sqrt{\max\{mse^*(\hat{\theta}, \hat{\mu}_s^{EB}), 0\}}, \quad (28)$$

$$I_2^{CEB*}(\hat{\theta}): \hat{\mu}_s^{EB}(\hat{\theta}) \pm z_{\alpha/2} \left[1 + h_2^*(\hat{\theta}) \right] \sqrt{g_1(\hat{\theta}) + g_2(\hat{\theta})}, \quad (29)$$

where $mse^*(\hat{\theta}, \hat{\mu}_s^{EB})$ is given in (24) and $h_1^*(\hat{\theta}) = (1 + z_{\alpha/2}^2)\{8g_1(\hat{\theta})^2\}^{-1} E_*[\{g_1(\hat{\theta}^*) - g_1(\hat{\theta})\}^2|\mathbf{y}]$ and $h_2^*(\hat{\theta}) = h_1^*(\hat{\theta}) + \{g_1(\hat{\theta}) - E_*[g_1(\hat{\theta}^*)|\mathbf{y}] + \bar{g}_3^*(\hat{\theta})\}/\{2g_1(\hat{\theta})\}$. As a confidence interval based on the Taylor series expansion, we treat the confidence interval with the correction term, given by

$$I^{CEB}(\hat{\theta}): \hat{\mu}_s^{EB}(\hat{\theta}) \pm z_{\alpha/2} \left[1 + h(\hat{\theta}) \right] \sqrt{mse(\hat{\theta}, \hat{\mu}_s^{EB})}, \quad (30)$$

where $mse(\hat{\theta}, \hat{\mu}_s^{EB})$ is given in (26) and $h(\theta) = (z_{\alpha/2}^2 + 1)\{8\theta^2(\theta + d_s)^2\}^{-1} d_s^2 \text{Var}(\hat{\theta}^\dagger)$. The confidence interval proposed by Chatterjee et al. [6] is different from ours. As seen from (6), the conditional distribution of μ_s given $\mathbf{y}$ is $\mu_s|\mathbf{y} \sim \mathcal{N}(\hat{\mu}_s(\mathbf{y}_s, \theta, \boldsymbol{\beta}), \sigma_s^2(\theta))$, where $\mu_s = \mathbf{x}_s' \boldsymbol{\beta} + v_s$, $\hat{\mu}_s(\mathbf{y}_s, \theta, \boldsymbol{\beta}) = \mathbf{x}_s' \boldsymbol{\beta} + \theta(\theta + d_s)^{-1}(\mathbf{y}_s - \mathbf{x}_s' \boldsymbol{\beta})$ and $\sigma_s^2(\theta) = d_s\theta(\theta + d_s)^{-1}$. Thus, $\sigma_s(\theta)^{-1}\{\mu_s - \hat{\mu}_s(\mathbf{y}_s, \theta, \boldsymbol{\beta})\} \sim \mathcal{N}(0, 1)$. Although this suggests to construct a confidence interval from the distribution of $\sigma_s(\hat{\theta})^{-1}\{\mu_s - \hat{\mu}_s(\mathbf{y}_s, \hat{\theta}, \hat{\boldsymbol{\beta}}(\hat{\theta}))\}$, the distribution is not normal. Define $\mathcal{L}(z)$ and $\mathcal{L}^*(z)$ by

$$\begin{aligned} \mathcal{L}(z) &= P \left[\sigma_s(\hat{\theta})^{-1} \{\mu_s - \hat{\mu}_s(\mathbf{y}_s, \hat{\theta}, \hat{\boldsymbol{\beta}}(\hat{\theta}))\} \leq z \right], \\ \mathcal{L}^*(z) &= P \left[\sigma_s(\hat{\theta}^*)^{-1} \{\mu_s^* - \hat{\mu}_s(\mathbf{y}_s^*, \hat{\theta}^*, \hat{\boldsymbol{\beta}}^*(\hat{\theta}^*))\} \leq z \right], \end{aligned}$$

where $\mu_s^* = \mathbf{x}_s' \hat{\boldsymbol{\beta}}(\hat{\theta}) + v_s^*$. Chatterjee et al. [6] proved that $\mathcal{L}(z)$ can be approximated by $\mathcal{L}^*(z)$ with the second-order accuracy, and proposed the confidence interval of μ_s given by

$$I^{CLL} = \left[\hat{\mu}_s(\mathbf{y}_s, \hat{\theta}, \hat{\boldsymbol{\beta}}(\hat{\theta})) - q_1 \sigma_s(\hat{\theta}), \hat{\mu}_s(\mathbf{y}_s, \hat{\theta}, \hat{\boldsymbol{\beta}}(\hat{\theta})) - q_2 \sigma_s(\hat{\theta}) \right], \quad (31)$$

where q_1 and q_2 satisfies that $\mathcal{L}^*(q_2) - \mathcal{L}^*(q_1) = 1 - \alpha$.

[3] AIC. Two kinds of Akaike Information Criteria based on the parametric bootstrap method are given by

$$\begin{aligned} AIC_1^* &= -2 \log f_m(\mathbf{y}|\hat{\boldsymbol{\beta}}(\hat{\theta}), \hat{\theta}) + 2p - \Delta_2^*(\mathbf{y}), \\ AIC_2^* &= -2 \log f_m(\mathbf{y}|\hat{\boldsymbol{\beta}}(\hat{\theta}), \hat{\theta}) - \Delta_1^*(\mathbf{y}) - \Delta_2^*(\mathbf{y}), \end{aligned} \quad (32)$$

where $\Delta_1^*(\mathbf{y}) = -2E_*[\mathbf{u}'\mathbf{P}(\hat{\theta}^*)\mathbf{u}^*|\mathbf{y}]$ and $\Delta_2^*(\mathbf{y}) = E_*[\mathbf{u}'\hat{\boldsymbol{\Sigma}}^{*-1}\mathbf{u}^* - \text{tr}[\hat{\boldsymbol{\Sigma}}\hat{\boldsymbol{\Sigma}}^{*-1}|\mathbf{y}]]$ for $\hat{\boldsymbol{\Sigma}} = \boldsymbol{\Sigma}(\hat{\theta})$, $\hat{\boldsymbol{\Sigma}}^* = \boldsymbol{\Sigma}(\hat{\theta}^*)$ and $\mathbf{u}^* = \mathbf{y}^* - \mathbf{X}\hat{\boldsymbol{\beta}}(\hat{\theta})$. On the other hand, AIC based on the Taylor expansion is described as

$$AIC = -2 \log f_m(\mathbf{y}|\hat{\boldsymbol{\beta}}(\hat{\theta}), \hat{\theta}) - \Delta(\hat{\theta}), \quad (33)$$

where $\Delta(\theta) = -2p - E[\text{tr}[\nabla_{\mathbf{y}}\nabla_{\mathbf{y}}'\hat{\theta}^\dagger]]$. This was derived by Kubokawa [21].

[4] Conditional AIC. The conditional AIC based on the parametric bootstrap method is given by

$$cAIC^* = -2 \log f(\mathbf{y}|\hat{\mathbf{v}}(\hat{\theta}), \hat{\boldsymbol{\beta}}(\hat{\theta}), \hat{\theta}) - \Delta_{c1}^*(\mathbf{y}) - \Delta_{c2}^*(\mathbf{y}), \quad (34)$$

where $\Delta_{c1}^*(\mathbf{y}) = -2\{E_*[\mathbf{u}'\hat{\boldsymbol{\Sigma}}^{-1}\mathbf{D}\mathbf{P}(\hat{\theta}^*)\mathbf{u}^* + \text{tr}[\mathbf{D}\hat{\boldsymbol{\Sigma}}^{*-1}|\mathbf{y}]] + k - 2\text{tr}[\mathbf{D}\hat{\boldsymbol{\Sigma}}^{-1}]\}$ and $\Delta_{c2}^*(\mathbf{y}) = 2E_*[\mathbf{u}'\hat{\boldsymbol{\Sigma}}^{-1}\mathbf{D}\hat{\boldsymbol{\Sigma}}^{*-1}\mathbf{u}^*|\mathbf{y}] - 2\text{tr}[\mathbf{D}\hat{\boldsymbol{\Sigma}}^{-1}]$. Kubokawa [21] derived the conditional AIC based on the Taylor series expansion given as

$$cAIC = -2 \log f(\mathbf{y}|\hat{\mathbf{v}}(\hat{\theta}), \hat{\boldsymbol{\beta}}(\hat{\theta}), \hat{\theta}) - \Delta_c(\hat{\theta}), \quad (35)$$

where $\Delta_c(\theta) = -2\rho(\theta) - 2\text{tr}[\mathbf{D}\boldsymbol{\Sigma}^{-1}E[\nabla_{\mathbf{y}}\nabla_{\mathbf{y}}'\hat{\theta}^\dagger]]$.

[5] Estimation of θ . As an estimator of θ , in this paper, we use the truncated Fay–Herriot estimator $\hat{\theta} = \max\{\theta_0, k^{-1}\}$, where θ_0 is the solution of the equation $L^{FH}(\theta_0) = 0$, where $L^{FH}(\theta_0) = \mathbf{y}'(\boldsymbol{\Sigma}(\theta_0)^{-1} - \mathbf{P}(\theta_0))\mathbf{y} - (k - p)$. For the truncated Fay–Herriot estimator, $B(\theta) = 2\{k\text{tr}[\boldsymbol{\Sigma}^{-2}] - (\text{tr}[\boldsymbol{\Sigma}^{-1}])^2/(\text{tr}[\boldsymbol{\Sigma}^{-1}])^3\}$, $V(\theta) = 2k/(\text{tr}[\boldsymbol{\Sigma}^{-1}])^2$, $\nabla_{\mathbf{y}}\hat{\theta} = 2\boldsymbol{\Sigma}^{-1}\mathbf{y}/\mathbf{y}'\boldsymbol{\Sigma}^{-2}\mathbf{y}/(\hat{\theta} > k^{-1})$, $\Delta(\theta) = -2(p + 1)$ and $\Delta_c(\theta) = -2\rho(\theta) - 2\text{tr}[\boldsymbol{\Sigma}^{-2}\mathbf{D}]/\text{tr}[\boldsymbol{\Sigma}^{-1}]$. For other estimators including ML, REML, Prasad–Rao and modified Fay–Herriot estimators, see [20] who summarized their biases and variances up to second order.

3.2. Simulation results

We investigate the performances of the proposed procedures by simulation and compare them with some existing procedures given in the literature. For the purpose, we adopt part of the simulation framework of [12] for our study. We consider the Fay–Herriot model (22) with $k = 15$, $\theta = 1$ and two d_i -patterns: (a) 0.7, 0.6, 0.5, 0.4, 0.3; (b) 4.0, 0.6, 0.5, 0.4, 0.1, which correspond to patterns (a) and (c) of [12]. Pattern (a) is less variable in d_i -values, while pattern (b) has larger variability. There are five groups $G_1, \dots, G_5$ and three small areas in each group. The sampling variances d_i are the same for area within the same group.

[Simulation experiment I]. Let us consider the case that $\mathbf{x}'_a\boldsymbol{\beta} = 0$ for simplicity as handled in [6]. Then, $\mu_s = v_s$, $\hat{\mu}_s^{EB} = \{1 - \gamma_s(\hat{\theta})\}y_s$ and $MSE(\theta, \hat{\mu}_s(\hat{\theta})) = d_s - d_s\gamma_s(\theta) + E[\{\gamma_s(\hat{\theta}) - \gamma_s(\theta)\}^2 y_s^2]$ for $\gamma_s(\theta) = d_s/(\theta + d_s)$. We prepare the true values of $MSE(\theta, \hat{\mu}_s(\hat{\theta}))$ in advance, which can be computed based on 100,000 simulated data. The relative bias and the risk functions of MSE estimator mse_s are given by

$$\begin{aligned} B_s(\theta, mse_s) &= E[mse_s - MSE(\theta, \hat{\mu}_s(\hat{\theta}))]/MSE(\theta, \hat{\mu}_s(\hat{\theta})), \\ R_s(\theta, mse_s) &= E[\{mse_s - MSE(\theta, \hat{\mu}_s(\hat{\theta}))\}^2]/\{MSE(\theta, \hat{\mu}_s(\hat{\theta}))\}^2. \end{aligned}$$

For confidence interval CI_s of $\mu_s = v_s$, the coverage probability and the expected length of CI_s are given by

$$\begin{aligned} CP_s(\theta, CI_s) &= P[\mu_s \in CI_s], \\ EL_s(\theta, CI_s) &= E[\text{Length of } CI_s]. \end{aligned}$$

These values are computed as average values based on 10,000 simulation runs where the size of the bootstrap sample is 1000. Further, those values are averaged over areas within groups G_i , $i = 1, \dots, 5$. The simulation results are generated using Ox version 5.10 (see [13]) and the Arfima package version 1.00 [14].

Concerning the MSE estimation, we handle the four estimators mse , mse^* , mse^{BL} and mse^E given in (24)–(27), which are referred as TLap, PBap, PBbl and ExactU, respectively. The values of their relative biases $10^2 \times B_s(\theta, mse_s)$ and risks $10^2 \times R_s(\theta, mse_s)$ by simulation are reported in Table 1. Since the ExactU is an unbiased estimator, it is clear that the values of the bias of ExactU are small, but it has very large risks. This means that estimating the MSE unbiasedly does not necessarily lead to improvement of the risk, but rather yields large variability in general. Investigating the relative biases and risks of TLap, PBap and PBbl in details, we can see that those values of TLap, PBap and PBbl are reasonably small. In comparison of their relative risks, it is seen that TLap has a slightly smaller risk than PBap, which has also a little bit smaller risk than PBbl,

Table 1Values of relative biases and risks of the four MSE estimators for $\theta = 1$.

			Bias $10^2 \times B(\theta, mse)$				Risk $10^2 \times R(\theta, mse)$			
	d_i	MSE	TLap	PBap	PBbl	ExactU	TLap	PBap	PBbl	ExactU
Pattern (a)										
G_1	0.7	0.441	−0.28	−0.20	−0.24	0.37	4.3	6.8	8.0	140.0
G_2	0.6	0.402	−0.09	0.36	0.22	−1.02	3.4	6.0	7.2	108.3
G_3	0.5	0.357	0.12	1.06	1.06	−0.13	2.5	5.1	6.5	86.8
G_4	0.4	0.305	0.40	1.96	2.00	−0.56	1.5	4.2	5.7	63.6
G_5	0.3	0.246	0.89	3.25	3.52	−0.54	0.6	3.3	5.2	43.5
Pattern (b)										
G_1	4.0	0.854	−1.69	−2.57	−2.16	1.97	22.3	22.7	24.9	2785.6
G_2	0.6	0.403	−0.38	0.25	0.09	0.26	4.6	6.6	7.9	115.3
G_3	0.5	0.358	−0.17	0.93	0.86	−0.37	3.5	5.6	7.0	87.7
G_4	0.4	0.306	0.07	1.82	1.92	−0.13	2.4	4.5	6.2	64.9
G_5	0.1	0.094	2.10	6.86	8.08	−0.11	0.2	1.6	4.6	11.3

Table 2Values of coverage probability and expected length of the four confidence intervals with nominal confidence coefficient 95% for $\theta = 1$.

		Cov. prob. $10^2 \times CP$				Expected length EL			
	d_i	TLap	PB_1	PB_2	PBcII	TLap	PB_1	PB_2	PBcII
Pattern (a)									
G_1	0.7	96.2	95.0	95.2	95.4	2.85	2.72	2.73	2.82
G_2	0.6	96.3	95.2	95.4	95.5	2.72	2.59	2.60	2.68
G_3	0.5	96.2	95.2	95.4	95.5	2.56	2.44	2.45	2.51
G_4	0.4	96.1	95.3	95.4	95.3	2.36	2.26	2.27	2.31
G_5	0.3	96.2	95.6	95.7	95.5	2.11	2.03	2.03	2.05
Pattern (b)									
G_1	4.0	96.2	94.7	95.8	95.2	4.05	3.95	3.99	4.15
G_2	0.6	96.1	95.1	95.3	95.2	2.66	2.60	2.61	2.67
G_3	0.5	96.1	95.1	95.3	95.2	2.50	2.45	2.45	2.50
G_4	0.4	96.2	95.4	95.5	95.4	2.30	2.26	2.27	2.30
G_5	0.1	95.6	95.8	95.9	95.3	1.25	1.26	1.26	1.23

but their differences are little. Thus, in this simulation experiment, the estimators PBap and PBbl based on the parametric bootstrap method are as good as TLap in both patterns (a) and (b).

Concerning the interval estimation, we handle the four confidence intervals I^{CEB} , I_1^{CEB*} , I_2^{CEB*} and I^{CLL} given in (28)–(31), which are referred as TLap, PB_1 , PB_2 and PBcll, respectively. The values of their coverage probabilities $10^2 \times CP$ and expected lengths EL by simulation are reported in Table 2. In both patterns, the four confidence intervals satisfy the nominal confidence level, but TLap has a tendency to take slightly larger coverage probability than 95%. It is also seen that PB_1 and PB_2 have smaller expected lengths than TLap and PBcll except G_5 in pattern (b). The difference between PB_1 and PB_2 appears in the coverage probability at G_1 in pattern (b). In fact, mse^* sometimes takes negative values in this case, and the resulting confidence interval PB_1 cannot construct an interval, which yields 94.7% coverage probability at G_1 , slightly smaller than the nominal confidence coefficient. PB_2 and PBcll are free from such a drawback. Thus, these observations show that the confidence interval PB_2 has a good performance in this simulation experiment.

Although simulation results are not reported here, we have investigated the performances of the parametric bootstrap procedures PB_1 , PB_2 and PBcll with the Prasad–Rao estimator for θ . The simulation results show that their performances are not bad in pattern (a), but for pattern (b), their coverage probabilities are much smaller than 95%. As defined in (23), the conditional distribution of y_a^* given $\mathbf{y}$ is $\mathcal{N}(\mathbf{x}_a^* \hat{\mathbf{B}}(\hat{\theta}), \hat{\theta} + d_a)$, and the variability of y^* strongly depends on the estimate $\hat{\theta}$. This implies that we need to use an estimator of θ with higher precision. Thus, in this simulation experiment, we employ the Fay–Herriot estimator instead of the Prasad–Rao estimator.

Since [6] used 1000 bootstrap sample size, we have treated the same sample size. It may be interesting to investigate what happens in the case of 100 bootstrap sample size. Although the simulation result is omitted here, it tells us that in the case of 100 bootstrap sample size, the coverage probability of PBcll is lower than 95%, but PB_1 and PB_2 still satisfies the nominal level. This suggests that 1000 bootstrap sample size is necessary for PBcll, but PB_1 and PB_2 work well in the simulation experiment with 100 bootstrap sample size.

[Simulation experiment II]. We next investigate the performances of AIC and conditional AIC derived in the previous sections through simulation and compare them in terms of the relative frequencies of selecting the true model.

Table 3

Relative frequencies (%) of candidate models selected by the five criteria AIC , AIC_1^* , AIC_2^* , $cAIC$ and $cAIC^*$ for $k = 15$ and $\theta = 1$: the dimension of a full model is $p = 7$ and the true model is (3).

(m)	Pattern (a)					Pattern (b)				
	AIC	AIC_1^*	AIC_2^*	$cAIC$	$cAIC^*$	AIC	AIC_1^*	AIC_2^*	$cAIC$	$cAIC^*$
(1)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.6
(2)	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.3	3.2
(3)	61.7	59.8	65.2	68.1	75.6	59.0	57.0	64.8	62.5	66.2
(4)	13.2	13.3	12.0	13.0	10.3	13.0	13.3	11.2	13.7	11.5
(5)	9.2	9.4	7.8	7.9	5.1	10.0	10.3	8.1	9.8	6.5
(6)	7.2	7.7	6.5	5.5	3.8	8.4	8.8	7.1	7.0	5.0
(7)	8.6	9.7	8.2	5.2	5.0	9.4	10.3	8.6	6.5	5.7
(3) + (4)	74.9	73.1	77.2	81.1	85.9	72.0	70.3	76.0	76.2	77.7

Let $\mathbf{x}_1, \dots, \mathbf{x}_k$ be generated from $\mathcal{N}_p(\mathbf{0}, \Sigma_x)$ where $\Sigma_x = (1 - \rho_x)\mathbf{I}_p + \rho_x \mathbf{J}_p$ for $\rho_x = 0.1$, where $\mathbf{J}_p = \mathbf{j}_p \mathbf{j}_p'$ for $\mathbf{j}_p = (1, \dots, 1)'$, a p -vector of ones. In this experiment, we consider a class of the nested models denoted by (m), which is described by

$$(m) \mathbf{y} = \mathbf{X}\boldsymbol{\beta}^{(m)} + \mathbf{Z}\mathbf{v} + \boldsymbol{\epsilon}, \quad \text{for } m = 1, \dots, p,$$

where $\boldsymbol{\beta}^{(m)} = (\beta_1, \dots, \beta_m, 0, \dots, 0)'$, and $\mathbf{v}$ and $\boldsymbol{\epsilon}$ are mutually independent random variables having $\mathbf{v} \sim \mathcal{N}_k(\mathbf{0}, \theta \mathbf{I})$ and $\boldsymbol{\epsilon} \sim \mathcal{N}_k(\mathbf{0}, \mathbf{D})$. It is noted that the first m components of $\boldsymbol{\beta}^{(m)}$ are not zero and the others are zero. Suppose that the true model is given by

$$(p^*) \mathbf{y} = \mathbf{X}\boldsymbol{\beta}^* + \mathbf{v} + \boldsymbol{\epsilon},$$

where $1 \leq p^* \leq p$ and $\boldsymbol{\beta}^* = (\beta_1, \dots, \beta_{p^*}, 0, \dots, 0)'$ for $\beta_\ell = 2(-1)^{\ell+1}\{1 + 1.3^{\ell+1}\}$, $1 \leq \ell \leq p^*$. We here handle the case that $\theta = 1$, $p = 7$, $p^* = 3$ and $k = 15$. Thus, the true model is (3), and the candidate models are (1), $\dots$, (7). Corresponding to the model (m), we generate random variables from the parametric bootstrap model $\mathbf{y}^* = \mathbf{X}\hat{\boldsymbol{\beta}}^{(m)} + \mathbf{Z}\mathbf{v}^* + \boldsymbol{\epsilon}^*$, where $\mathbf{v}^*$ and $\boldsymbol{\epsilon}^*$ are defined below (23).

We compare selection criteria in the sense of relative frequency of selecting the true model (3). The criteria we examine are the existing procedures AIC and $cAIC$ based on the Taylor series expansion, and the proposed ones AIC_1^* , AIC_2^* and $cAIC^*$ based on the parametric bootstrap method. For each criterion and each candidate model (m), the number of selecting the model (m) is counted for 10,000 data set. We thus obtain the relative frequencies of the model (m) selected by the criteria by dividing the number by 10,000. These relative frequencies are reported in Table 3, where the last column denotes the sum of two relative frequencies of (3) and (4). Although the model (4) is not true, it includes the true model (3), so that it may be not bad to look at the sum of the two relative frequencies. From Table 3, it is seen that AIC , AIC_1^* and AIC_2^* have similar performances, and AIC_2^* is slightly better than AIC and AIC_1^* . Concerning the conditional AIC , $cAIC^*$ performs similarly to $cAIC$, and $cAIC^*$ is better than $cAIC$. In this simulation experiment, it is seen that among the five criteria, $cAIC^*$ is superior in light of relative frequency of selecting the true model.

3.3. Example of posted land price data

We apply the proposed procedures to the posted land price data along the Keikyu train line which connects the suburbs in Kanagawa prefecture to the Tokyo metropolitan area. This data set was used by Kubokawa [19] who gave confidence intervals based on EBLUP.

A data set of the posted land price data in 2001 and their covariates are available for 48 stations on the Keikyu train line, and we consider each station as a small area, namely, $k = 48$. For the a th station, there are data of n_a land spots, where the average of n_a 's is 3.73. Since those who live in the suburbs take this line to work or study in Tokyo on weekdays, it may be expected that the land price depends on the distance from Tokyo. For $b = 1, \dots, n_a$, we use five kinds of observations y_{ab} , TRN_a , DST_{ab} , $FOOT_{ab}$ and FAR_{ab} , where y_{ab} denotes the value of the posted land price (Yen in hundred of thousands) per m^2 of the b th spot, TRN_a is the time to take by train from the station a to the Tokyo station around 8:30 in the morning, DST_{ab} is the geographical distance from the spot b to the nearby station a , $FOOT_{ab}$ is the time to take on foot from the spot b to the nearby station a and FAR_{ab} denotes the floor-area ratio of the spot b . As regressor variables, we consider nine variables FAR_{ab} , TRN_a , TRN_a^2 , DST_{ab} , DST_{ab}^2 , $FOOT_{ab}$, $FOOT_{ab}^2$, $TRN_a \times DST_{ab}$ and $TRN_a \times FOOT_{ab}$, which are denoted by $x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8$ and x_9 . Also, the constant term is denoted by x_0 .

To analyze the data, we employ the nested error regression model (NERM), which is described as

$$y_{ab} = \mathbf{x}_{ab}'\boldsymbol{\beta} + v_a + \varepsilon_{ab}, \quad a = 1, \dots, k, \quad b = 1, \dots, n_a, \quad (36)$$

where $\mathbf{x}_{ab}'$ is a vector of values of the explanatory variables $(x_0, x_1, \dots, x_9)$, $N = \sum_{a=1}^k n_a$, and v_a 's and ε_{ab} 's are mutually independently distributed as $v_a \sim \mathcal{N}(0, \sigma_v^2)$ and $\varepsilon_{ab} \sim \mathcal{N}(0, \sigma^2)$ for unknown variance components σ_v^2 and σ^2 . For the unknown variance components, we here use the Prasad–Rao estimators given by Prasad and Rao [25]. We consider nested error regression model (36) with the regressors x_0 – x_9 as a full model.

Table 4

AIC and conditional AIC for selecting regressor variables in the posted land price data.

m	x_i	AIC_1^*	$cAIC^*$	$\hat{\sigma}^{2U}$	$\hat{\sigma}_v^{2TR}$
01	x_0	566	512	0.796	1.188
02	x_0, x_1	461	415	0.449	0.481
03	x_0, x_1, x_2	407	396	0.452	0.131
04	x_0, x_1, x_2, x_3	395	389	0.456	0.079
05	x_0, x_1, x_2, x_3, x_4	395	390	0.459	0.071
06	$x_0, x_1, x_2, x_3, x_4, x_9$	398	393	0.462	0.070
07	$x_0, x_1, x_2, x_3, x_4, x_8, x_9$	465	425	0.466	0.371
10	All x_i 's	489	451	0.470	0.416

Table 5

Examples of values of EBLUP, MSE estimates and confidence intervals for the average land price (Yen in thousands).

s	n_a	Prediction					Confidence interval		
		$\bar{y}_s$	$\bar{x}_a' \hat{\beta}$	$\hat{\mu}_s^{EB}$	$\sqrt{mse_s^*}$	$\hat{\sigma} / \sqrt{n_a}$	Lower	Upper	Length
08	02	401	426	420	28	47	356	484	128
16	02	285	317	309	27	47	249	369	121
24	03	312	350	338	26	39	280	395	114
32	12	247	177	223	19	19	181	265	084
40	03	285	270	275	26	39	216	333	117
48	04	159	203	186	28	33	120	252	132

Table 4 reports values of AIC_1^* and $cAIC^*$ given in (18) and (21) for several candidate models, where the regressors which minimizes the information criteria are added to the model based on the forward stepwise selection. Among these candidate models, AIC_1^* selects both $\{x_0, x_1, x_2, x_3\}$ and $\{x_0, x_1, x_2, x_3, x_4\}$. Although the minimum of $cAIC^*$ is attained at the model with $\{x_0, x_1, x_2, x_3\}$, we can select the variables $\{x_0, x_1, x_2, x_3, x_4\}$ since their difference is not significant. Since it may be better to explain the model including more regressor variables, we suggest the model with $\{x_0, x_1, x_2, x_3, x_4\}$, namely,

$$y_{ab} = \beta_0 + FAR_a \beta_1 + TRN_a \beta_2 + (TRN_a)^2 \beta_3 + FOOT_a \beta_4 + v_a + \varepsilon_{ab},$$

for $a = 1, \dots, k$ and $b = 1, \dots, n_a$. The parameters are estimated by $\hat{\sigma}^2 = 0.45936$, $\hat{\sigma}_v^2 = 0.07154$ and $(\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3, \hat{\beta}_4) = (5.1288, 6.3937 \times 10^{-3}, -0.1076, 7.0710 \times 10^{-4}, -8.1562 \times 10^{-5})$. This result demonstrates that the land prices are not only decreasing as a quadratic function of TRN_a , time to take from the nearby station to Tokyo station, but also decreasing in $FOOT_{ab}$, time to take from the land spot to the nearby station.

We now estimate the average land price per m^2 around the a th station, namely, $\mu_a = \bar{x}_a' \beta + v_a$ for $a = 1, \dots, 48$ where $\bar{x}_a$ is the mean of the regressor variables selected above. For example, for the six stations 8, 16, 24, 32, 40 and 48, the values of $n_a, \bar{y}_a, \bar{x}_a' \hat{\beta}$ and EBLUP $\hat{\mu}_a^{EB}$ are reported in the column of 'prediction' in Table 5, where all the values are given in Yen in thousands. Also, the table reports the values of $\sqrt{mse_a^*}$, which is a square root of estimates of MSE given in (10), and $\hat{\sigma} / \sqrt{n_a}$, which is a square root of estimates of the conditional variance of $\bar{y}_a$ given v_a . The lower and upper end-points of the confidence interval I_2^{CEB*} , given in (14), with 95% confidence coefficient and the length of the interval are reported in the column of 'confidence interval' in Table 5. From Table 5, it is revealed that for smaller n_a , the EBLUP $\hat{\mu}_a^{EB}$ shrinks $\bar{y}_a$ much more toward $\bar{x}_a' \hat{\beta}$, which results in smaller values $\sqrt{mse_a^*}$ than $\hat{\sigma} / \sqrt{n_a}$. For example, $\sqrt{mse_a^*}$ is about half of $\hat{\sigma} / \sqrt{n_a}$ for $n_a = 1$, but $\sqrt{mse_a^*}$ is equal to $\hat{\sigma} / \sqrt{n_a}$ for $n_a = 12$. Also, it is seen that the confidence intervals give shorter intervals for larger n_a .

4. Concluding remarks

In this paper, we have suggested the procedures based on the parametric bootstrap methods in the estimation of the MSE of EBLUP, the confidence interval based on EBLUP, and variable selection problems based on AIC and conditional AIC. These procedures are not only easy to implement practically, but also justified theoretically to have second-order approximations. Their performances have been investigated through simulation study.

We have used the Fay–Herriot model to compare the proposed bootstrap methods with the existing procedures. However, it would be interesting to treat more complicated models like the model for combining cross-sectional and time-series data proposed by Rao and Yu [27]. The application to such complicated models will be studied in a future.

Acknowledgments

The authors are grateful to the Associate Editor and two referees for their valuable comments and helpful suggestions. This research was supported in part by Grant-in-Aid for Scientific Research Nos. 19200020 and 21540114 from Japan Society for the Promotion of Science.

Appendix

We here give second-order justifications of the procedures proposed in Section 2.

A.1. Notations and common assumptions

We begin by introducing the notations used here. Let $\mathcal{C}_\theta^{[k]}$ denote a set of k times continuously differentiable functions with respect to θ . For partial derivatives with respect to θ , we utilize the notations

$$\begin{aligned} \mathbf{A}_{(i)}(\theta) &= \partial_i \mathbf{A}(\theta) = \frac{\partial \mathbf{A}(\theta)}{\partial \theta_i}, & \mathbf{A}_{(ij)}(\theta) &= \partial_{ij} \mathbf{A}(\theta) = \frac{\partial^2 \mathbf{A}(\theta)}{\partial \theta_i \partial \theta_j}, \\ \mathbf{A}_{(ijk)}(\theta) &= \partial_{ijk} \mathbf{A}(\theta) = \frac{\partial^3 \mathbf{A}(\theta)}{\partial \theta_i \partial \theta_j \partial \theta_k}, \end{aligned}$$

where $\mathbf{A}(\theta)$ is a scalar, vector or matrix. For the first and second differential operators with respect to $\mathbf{y}$, we use the notations

$$\nabla_{\mathbf{y}} = \frac{\partial}{\partial \mathbf{y}}, \quad \nabla_{\mathbf{y}} \nabla'_{\mathbf{y}} = \frac{\partial}{\partial \mathbf{y}} \frac{\partial}{\partial \mathbf{y}'},$$

namely, the i -th element of $\nabla_{\mathbf{y}}$ and the (i, j) -th element of $\nabla_{\mathbf{y}} \nabla'_{\mathbf{y}}$ are $\partial/\partial y_i$ and $\partial^2/\partial y_i \partial y_j$, respectively.

For $0 \leq i, j, k \leq q$, let $\lambda_1(\Sigma) \leq \dots \leq \lambda_N(\Sigma)$ be the eigenvalues of Σ and let those of $\Sigma_{(i)}$, $\Sigma_{(ij)}$ and $\Sigma_{(ijk)}$ be $\lambda_a^i(\Sigma)$, $\lambda_a^{ij}(\Sigma)$ and $\lambda_a^{ijk}(\Sigma)$ for $a = 1, \dots, N$ respectively, where $|\lambda_1^i(\Sigma)| \leq \dots \leq |\lambda_N^i(\Sigma)|$, $|\lambda_1^{ij}(\Sigma)| \leq \dots \leq |\lambda_N^{ij}(\Sigma)|$ and $|\lambda_1^{ijk}(\Sigma)| \leq \dots \leq |\lambda_N^{ijk}(\Sigma)|$.

Throughout the paper, assume the following conditions for large N and $0 \leq i, j, k \leq q$:

(A1) The elements of $\mathbf{X}, \mathbf{Z}, \mathbf{G}(\theta), \mathbf{R}(\theta), p$ and q are bounded, and $\mathbf{X}'\mathbf{X}$ is positive definite and $\mathbf{X}'\mathbf{X}/N$ converges to a positive definite matrix;

(A2) $\Sigma(\theta) \in \mathcal{C}_\theta^{[3]}$, $\lim_{N \rightarrow \infty} \lambda_1(\Sigma) > 0$, $\lim_{N \rightarrow \infty} \lambda_N(\Sigma) < \infty$, $\lim_{N \rightarrow \infty} |\lambda_N^i(\Sigma)| < \infty$, $\lim_{N \rightarrow \infty} |\lambda_N^{ij}(\Sigma)| < \infty$ and $\lim_{N \rightarrow \infty} |\lambda_N^{ijk}(\Sigma)| < \infty$.

(A3) $\hat{\theta} = \hat{\theta}(\mathbf{y})$ is an estimator of θ which satisfies that $\hat{\theta}(-\mathbf{y}) = \hat{\theta}(\mathbf{y})$ and $\hat{\theta}(\mathbf{y} + \mathbf{X}\alpha) = \hat{\theta}(\mathbf{y})$ for any p -dimensional vector α .

(A4) It is assumed that $\hat{\theta} - \theta$ is expanded as

$$\hat{\theta} - \theta = \hat{\theta}^\dagger + \hat{\theta}^{\dagger\dagger} + O_p(N^{-3/2}), \quad (\text{A.1})$$

where $\hat{\theta}^\dagger = O_p(N^{-1/2})$, $\hat{\theta}^{\dagger\dagger} = O_p(N^{-1})$ and $E[\hat{\theta}^\dagger] = \mathbf{0}$.

As checked in [20], ML, REML, the Prasad–Rao estimator and the Fay–Herriot estimator satisfy the conditions (A1)–(A4) in the Fay–Herriot model and the nested error regression model.

A.2. MSE estimation and interval estimation

[1] *Second-order unbiasedness of the MSE estimator.* We first treat the estimation of MSE of the EBLUP $\hat{\mu}^{EB}(\hat{\theta})$, where $\hat{\mu}^{EB}(\theta) = \hat{\mu}^B(\beta(\theta), \theta) = \mathbf{a}'\beta(\theta) + \mathbf{s}(\theta)'(\mathbf{y} - \mathbf{X}\beta(\theta))$ for $\mathbf{s}(\theta) = \Sigma(\theta)^{-1}\mathbf{ZG}(\theta)\mathbf{b}$, and show that the MSE estimator $mse^*(\hat{\theta}, \hat{\mu}^{EB}(\hat{\theta}))$ given in (10) has the second-order unbiasedness.

Let $\bar{g}_3(\theta)$ and $\bar{g}_4(\theta)$ be functions defined by

$$\begin{aligned} \bar{g}_3(\theta) &= \text{tr} \left[\left(\frac{\partial \mathbf{s}(\theta)'}{\partial \theta} \right) \Sigma(\theta) \left(\frac{\partial \mathbf{s}(\theta)'}{\partial \theta} \right)' \mathbf{Cov}(\hat{\theta}^\dagger) \right], \\ \bar{g}_4(\theta) &= \left(\frac{\partial \mathbf{g}_1(\theta)}{\partial \theta} \right)' E[\hat{\theta}^{\dagger\dagger}] + \frac{1}{2} \text{tr} \left[\mathbf{B}(\theta) \mathbf{Cov}(\hat{\theta}^\dagger) \right] - \bar{g}_3(\theta), \end{aligned} \quad (\text{A.2})$$

where $\mathbf{Cov}(\hat{\theta}^\dagger) = E[(\hat{\theta}^\dagger)(\hat{\theta}^\dagger)']$, and the (i, j) -th element of $\mathbf{B}(\theta)$ is given by

$$(\mathbf{B}(\theta))_{ij} = (\mathbf{b} - \mathbf{Z}'\mathbf{s}(\theta))'(\partial_{ij}\mathbf{G}(\theta))(\mathbf{b} - \mathbf{Z}'\mathbf{s}(\theta)) + \mathbf{s}(\theta)'(\partial_{ij}\mathbf{R}(\theta))\mathbf{s}(\theta).$$

It is noted that $\mathbf{B}(\theta) = \mathbf{0}$ when $\mathbf{G}$ and $\mathbf{R}$ are matrices of linear functions of θ .

To establish the second-order approximation, we assume the following conditions:

(B1) The elements of $\mathbf{a}$ and $\mathbf{b}$ are uniformly bounded, and $\mathbf{s}(\theta)$ satisfies that $\mathbf{s}(\theta) \in \mathcal{C}_\theta^{[2]}$, $(\mathbf{y} - \mathbf{X}\beta)' \mathbf{s}(\theta) = O_p(1)$, $(\mathbf{y} - \mathbf{X}\beta)' \mathbf{s}_{(i)}(\theta) = O_p(1)$, $(\mathbf{y} - \mathbf{X}\beta)' \mathbf{s}_{(ij)}(\theta) = O_p(1)$, $\mathbf{s}_{(i)}(\theta)' \mathbf{s}_{(j)}(\theta) = O(1)$ and $\mathbf{s}_{(ij)}(\theta)' \Sigma(\theta) \nabla_{\mathbf{y}} \hat{\theta}_i^\dagger = O_p(N^{-1})$.

(B2) For $1 \leq i, j, k \leq q$, (i) $g_1(\theta) \in \mathcal{C}_\theta^{[3]}$ and $\partial_i g_1(\theta) = O(1)$, $\partial_{ij} g_1(\theta) = O(1)$ and $\partial_{ijk} g_1(\theta) = O(1)$, (ii) $g_2(\theta)$, $\bar{g}_3(\theta)$ and $\bar{g}_4(\theta)$ are continuously differentiable functions satisfying that $\partial_i g_2(\theta) = O(N^{-1})$, $\partial_i \bar{g}_3(\theta) = O(N^{-1})$ and $\partial_i \bar{g}_4(\theta) = O(N^{-1})$.

Theorem A.1. Assume the conditions (A1)–(A4) and (B1)–(B2). Then, $mse^*(\hat{\theta}, \hat{\mu}^{EB}(\hat{\theta}))$ is approximated as

$$mse^*(\hat{\theta}, \hat{\mu}^{EB}(\hat{\theta})) = mse(\hat{\theta}, \hat{\mu}^{EB}(\hat{\theta})) + O_p(N^{-3/2}), \quad (A.3)$$

where

$$mse(\hat{\theta}, \hat{\mu}^{EB}(\hat{\theta})) = g_1(\hat{\theta}) + g_2(\hat{\theta}) + \bar{g}_3(\hat{\theta}) - \bar{g}_4(\hat{\theta}). \quad (A.4)$$

Thus,

$$E[mse^*(\hat{\theta}, \hat{\mu}^{EB}(\hat{\theta}))] = MSE(\theta, \hat{\mu}^{EB}(\hat{\theta})) + O(N^{-3/2}). \quad (A.5)$$

Proof. Kubokawa [20] proved in his Theorems 2.1 and 2.2 that

$$g_3(\theta) = E[\{\hat{\mu}^{EB}(\hat{\theta}) - \hat{\mu}^{EB}(\theta)\}^2] = \bar{g}_3(\theta) + O(N^{-3/2}), \quad (A.6)$$

$$E[g_1(\hat{\theta})] = g_1(\theta) + \bar{g}_4(\theta) + O(N^{-3/2}), \quad (A.7)$$

which can be established under the conditions (A1)–(A4), (B1) and (B2)(i). Then from (A.7), it follows that

$$E_*[g_1(\hat{\theta}^*)|\mathbf{y}] = g_1(\hat{\theta}) + \bar{g}_4(\hat{\theta}) + O_p(N^{-3/2}). \quad (A.8)$$

Since $g_2(\theta) = O(N^{-1})$, it is seen that under condition (B2)(ii),

$$E_*[g_2(\hat{\theta}^*)|\mathbf{y}] = g_2(\hat{\theta}) + O_p(N^{-3/2}). \quad (A.9)$$

Hence from (A.8) to (A.9), it is seen that under condition (B2),

$$2\{g_1(\hat{\theta}) + g_2(\hat{\theta})\} - E_*[g_1(\hat{\theta}^*) + g_2(\hat{\theta}^*)|\mathbf{y}] = g_1(\hat{\theta}) - \bar{g}_4(\hat{\theta}) + g_2(\hat{\theta}) + O_p(N^{-3/2}). \quad (A.10)$$

Since $\mathbf{s}(\theta) = O(1)$, it is noted that

$$\mathbf{s}(\hat{\theta}) = \mathbf{s}(\theta) + \frac{\partial \mathbf{s}(\theta)}{\partial \theta'} \hat{\theta}^\dagger + O_p(N^{-1}),$$

which implies that

$$E[\{\mathbf{s}(\hat{\theta}) - \mathbf{s}(\theta)\}' \Sigma(\theta) \{\mathbf{s}(\hat{\theta}) - \mathbf{s}(\theta)\}] = \bar{g}_3(\theta) + O(N^{-3/2}).$$

Thus, we get

$$E_*[\{\mathbf{s}(\hat{\theta}^*) - \mathbf{s}(\hat{\theta})\}' \Sigma(\hat{\theta}) \{\mathbf{s}(\hat{\theta}^*) - \mathbf{s}(\hat{\theta})\}|\mathbf{y}] = \bar{g}_3(\hat{\theta}) + O_p(N^{-3/2}). \quad (A.11)$$

Combining (A.10) and (A.11) gives

$$\begin{aligned} mse^*(\hat{\theta}, \hat{\mu}^{EB}(\hat{\theta})) &= 2\{g_1(\hat{\theta}) + g_2(\hat{\theta})\} - E_*[g_1(\hat{\theta}^*) + g_2(\hat{\theta}^*)|\mathbf{y}] + E_*[\{\mathbf{s}(\hat{\theta}^*) - \mathbf{s}(\hat{\theta})\}' \Sigma(\hat{\theta}) \{\mathbf{s}(\hat{\theta}^*) - \mathbf{s}(\hat{\theta})\}|\mathbf{y}] \\ &= \{g_1(\hat{\theta}) - \bar{g}_4(\hat{\theta}) + g_2(\hat{\theta})\} + \bar{g}_3(\hat{\theta}) + O_p(N^{-3/2}) \\ &= mse(\hat{\theta}, \hat{\mu}^{EB}(\hat{\theta})) + O_p(N^{-3/2}). \end{aligned}$$

Finally, it follows from [20] that

$$E[mse^*(\hat{\theta}, \hat{\mu}^{EB}(\hat{\theta}))] = E[mse(\hat{\theta}, \hat{\mu}^{EB}(\hat{\theta}))] + O(N^{-3/2}),$$

which is equal to $MSE(\theta, \hat{\mu}^{EB}(\hat{\theta})) + O(N^{-3/2})$ and proves (A.5). $\square$

[2] *Second-order corrected confidence interval.* We next treat the interval estimation of μ with the corrected confidence intervals $I_1^{CEB*}(\hat{\theta})$ and $I_2^{CEB*}(\hat{\theta})$ given in (13) and (14), and show that the coverage probability can be approximated to the confidence coefficient in the second-order accuracy.

For the purpose, let us define function $h_1(\theta)$ by

$$h_1(\theta) = \frac{z_{\alpha/2}^2 + 1}{8g_1(\theta)^2} \text{tr} \left[\left(\frac{\partial g_1(\theta)}{\partial \theta} \right) \left(\frac{\partial g_1(\theta)}{\partial \theta} \right)' \text{Cov}(\hat{\theta}^*) \right]. \quad (A.12)$$

Assume the following condition:

(B3) $h_1(\theta) \in \mathcal{C}_\theta^{[1]}$ and $\partial_i h_1(\theta) = O(N^{-1})$ for $1 \leq i \leq q$.

Theorem A.2. Assume the conditions (A1)–(A4) and (B1)–(B3). Then,

$$P[\mu \in I_1^{CEB*}(\hat{\theta})] = 1 - \alpha + O(N^{-3/2}). \quad (\text{A.13})$$

Proof. Let us define U and V^* by

$$U = \{\hat{\mu}^{EB}(\hat{\theta}) - \hat{\mu}^B(\beta, \theta)\} / \sqrt{g_1(\theta)},$$

$$V^* = \left\{ [1 + h_1^*(\hat{\theta})] \sqrt{mse^*(\hat{\theta})} - \sqrt{g_1(\theta)} \right\} / \sqrt{g_1(\theta)},$$

where $\hat{\mu}^B(\beta, \theta) = \mathbf{a}'\beta + \mathbf{s}(\theta)'(\mathbf{y} - \mathbf{X}\beta)$. Using the same arguments as in the proof of Theorem 2.3 of [20], we can verify that

$$P[\mu \in I_1^{CEB*}(\hat{\theta})] = 1 - \alpha + \phi(z_\alpha)H_1^*(\theta) + O(N^{-3/2}), \quad (\text{A.14})$$

where

$$H_1^*(\theta) = z_{\alpha/2}E\left[2V^* - \{U^2 + z_{\alpha/2}^2(V^*)^2\}\right]$$

$$= z_{\alpha/2}E\left[\{2V^* + (V^*)^2\} - U^2 - (1 + z_{\alpha/2}^2)(V^*)^2\right]. \quad (\text{A.15})$$

It is noted that $mse^*(\hat{\theta}) = mse(\hat{\theta}) + O_p(N^{-3/2})$ and $h_1^*(\hat{\theta}) = h_1(\hat{\theta}) + O_p(N^{-3/2})$, where $mse(\hat{\theta})$ is given in (A.4). Then, it can be seen that

$$2V^* + (V^*)^2 = (2 + V^*)V^* = \{1 + h_1^*(\hat{\theta})\}^2 \frac{mse^*(\hat{\theta})}{g_1(\theta)} - 1$$

$$= \{1 + h_1(\hat{\theta})\}^2 \frac{mse(\hat{\theta})}{g_1(\theta)} - 1 + O_p(N^{-3/2})$$

$$= 2V + V^2 + O_p(N^{-3/2}),$$

where $V = [1 + h_1(\hat{\theta})] \sqrt{mse(\hat{\theta})} / \sqrt{g_1(\theta)} - 1$. Since $V^* = V + O_p(N^{-1})$, it is verified that $(V^*)^2 = V^2 + O_p(N^{-3/2})$. Hence, from (A.15), it follows that $H_1^*(\theta) = H_1(\theta) + O(N^{-3/2})$, where $H_1(\theta) = z_\alpha E[(2V + V^2) - U^2 - (1 + z_{\alpha/2}^2)V^2]$. This approximation is used to rewrite (A.14) as

$$P[\mu \in I_1^{CEB*}(\hat{\theta})] = 1 - \alpha + \phi(z_{\alpha/2})H_1(\theta) + O(N^{-3/2}). \quad (\text{A.16})$$

Kubokawa [20] showed in the proof of his Theorem 2.3 that $H_1(\theta) = O(N^{-3/2})$, which proves the theorem. $\square$

Theorem A.3. Under the same conditions as in Theorem A.2,

$$P[\mu \in I_2^{CEB*}(\hat{\theta})] = 1 - \alpha + O(N^{-3/2}). \quad (\text{A.17})$$

Proof. In the proof of Theorem A.2, we need to replace $mse^*(\hat{\theta})$, V^* and $h_1^*(\hat{\theta})$ with $g_1(\hat{\theta}) + g_2(\hat{\theta})$, W and $h_2^*(\hat{\theta})$, respectively, where $W = \{[1 + h_2^*(\hat{\theta})]\{g_1(\hat{\theta}) + g_2(\hat{\theta})\}^{1/2} - \sqrt{g_1(\theta)}\} / \sqrt{g_1(\theta)}$. Then,

$$P[\mu \in I_2^{CEB*}(\hat{\theta})] = 1 - \alpha + \phi(z_\alpha)H_2^*(\theta) + O(N^{-3/2}),$$

where

$$H_2^*(\theta) = z_{\alpha/2}E\left[\{2W + W^2\} - U^2 - (1 + z_{\alpha/2}^2)W^2\right].$$

Since $2W + W^2 = \{1 + h_2^*(\hat{\theta})\}^2 \{g_1(\hat{\theta}) + g_2(\hat{\theta})\} / g_1(\theta) - 1 = \{g_1(\hat{\theta}) + g_2(\hat{\theta})\} / g_1(\theta) - 1 + 2h_2^*(\hat{\theta})g_1(\hat{\theta}) / g_1(\theta) + O_p(N^{-3/2})$, it is seen that

$$E[2W + W^2] = \frac{E[g_1(\hat{\theta})] - g_1(\theta)}{g_1(\theta)} + \frac{g_2(\theta)}{g_1(\theta)} + 2E[h(\hat{\theta})] + O(N^{-3/2}).$$

It follows from the proof of Theorem 2.3 of [20] that $E[U^2] = \{g_2(\theta) + \bar{g}_3(\theta)\} / g_1(\theta) + O(N^{-3/2})$ for $\bar{g}_3(\theta)$ given in (A.2) and (A.6). Also it can be shown that $E[(1 + z_{\alpha/2}^2)W^2] = 2h_1(\theta) + O(N^{-3/2})$. Combining these approximations gives

$$H_2^*(\theta) = 2z_{\alpha/2} \left\{ h_2^*(\hat{\theta}) + \frac{E[g_1(\hat{\theta})] - g_1(\theta) - \bar{g}_3(\theta)}{2g_1(\theta)} - h_1(\theta) \right\}.$$

It is here noted that $E[E_*[g_1(\hat{\theta}^*)|\mathbf{y}] - g_1(\hat{\theta})] = E[g_1(\hat{\theta})] - g_1(\theta) + O(N^{-3/2})$, which can be verified from (A.7) to (A.8). Also note that $h_1^*(\hat{\theta}) = h_1(\theta) + O_p(N^{-3/2})$ and $\bar{g}_3^*(\hat{\theta}) = \bar{g}_3(\theta) + O_p(N^{-3/2})$. This implies that $H_2^*(\theta) = O(N^{-3/2})$, and the proof is complete. $\square$

A.3. Derivation of AIC_1^* , AIC_2^* and $cAIC^*$

We here show that the penalty terms of AIC_1^* , AIC_2^* and $cAIC^*$ given in (18) and (21) are second-order approximations of unbiased estimators of the corresponding biases.

Concerning AIC_1^* and AIC_2^* given in (18), let us define $c(\theta) = -\sum_{i=1}^q E[\text{tr}[\Sigma_{(i)} \nabla_{\mathbf{y}} \nabla_{\mathbf{y}}' \hat{\theta}_i^\dagger]]$ for $\hat{\theta}^\dagger = (\hat{\theta}_1^\dagger, \dots, \hat{\theta}_q^\dagger)'$. Assume the following conditions:

(C1) $\hat{\theta}^\dagger$ and $\hat{\theta}^{\dagger\dagger}$ satisfy that $E[\text{tr}[\nabla_{\mathbf{y}} \nabla_{\mathbf{y}}' \hat{\theta}_i^{\dagger\dagger}]] = O(N^{-1})$, $E[\text{tr}[(\nabla_{\mathbf{y}} \nabla_{\mathbf{y}}' \hat{\theta}_i^\dagger) \hat{\theta}_j^\dagger]] = O(N^{-1})$ and $E[\text{tr}[(\nabla_{\mathbf{y}} \hat{\theta}_i^\dagger)(\nabla_{\mathbf{y}} \hat{\theta}_j^\dagger)']] = O(N^{-1})$ for $\hat{\theta}^{\dagger\dagger} = (\hat{\theta}_1^{\dagger\dagger}, \dots, \hat{\theta}_q^{\dagger\dagger})'$.

(C2) $c(\theta) \in \mathcal{C}^{[1]}\theta$ and $\partial_i c(\theta) = O(N^{-1})$ for $1 \leq i \leq q$.

Theorem A.4. Assume the conditions (A1)–(A4) and (C1)–(C2). Then,

$$E[-2 \log f_m(\mathbf{y}|\hat{\beta}(\hat{\theta}), \hat{\theta}) - \Delta_1^*(\mathbf{y}) - \Delta_2^*(\mathbf{y})] = AI(\theta) + O(N^{-1/2})$$

for $\Delta_1^*(\mathbf{y})$ and $\Delta_2^*(\mathbf{y})$ given in (17). Also, $E[\Delta_1^*(\mathbf{y})] = -2p + O(N^{-1/2})$ and $E[\Delta_2^*(\mathbf{y})] = c(\theta) + O(N^{-1/2})$.

Proof. Define $\Delta_1(\theta)$ and $\Delta_2(\theta)$ by $\Delta_1(\theta) = -2E[(\mathbf{y} - \mathbf{X}\beta)' \mathbf{P}(\hat{\theta})(\mathbf{y} - \mathbf{X}\beta)]$ and $\Delta_2(\theta) = E[(\mathbf{y} - \mathbf{X}\beta)' \hat{\Sigma}^{-1}(\mathbf{y} - \mathbf{X}\beta) - \text{tr}[\Sigma \hat{\Sigma}^{-1}]]$. Kubokawa [21] proved in the proof of his Theorem 2.1 that

$$E_{\mathbf{y}}[-2 \log f_m(\mathbf{y}|\hat{\beta}(\hat{\theta}), \hat{\theta})] - AI(\theta) = \Delta_1(\theta) + \Delta_2(\theta),$$

and that $\Delta_1(\theta) = -2p + O(N^{-1/2})$ and $\Delta_2(\theta) = c(\theta) + O(N^{-1/2})$. Thus, it is observed that

$$E_*[(\mathbf{y}^* - \mathbf{X}\hat{\beta}(\hat{\theta}))' \mathbf{P}(\hat{\theta}^*)(\mathbf{y}^* - \mathbf{X}\hat{\beta}(\hat{\theta}))|\mathbf{y}] = p + O_p(N^{-1/2}),$$

$$E_*[(\mathbf{y}^* - \mathbf{X}\hat{\beta}(\hat{\theta}))' \Sigma(\hat{\theta}^*)^{-1}(\mathbf{y}^* - \mathbf{X}\hat{\beta}(\hat{\theta})) - \text{tr}[\Sigma(\hat{\theta})\Sigma(\hat{\theta}^*)^{-1}]|\mathbf{y}] = c(\hat{\theta}) + O_p(N^{-1/2}).$$

Since $c(\theta) = O(1)$, this implies that $E[\Delta_1^*(\mathbf{y})] = -2p + O(N^{-1/2})$ and $E[\Delta_2^*(\mathbf{y})] = c(\theta) + O(N^{-1/2})$, and we get the theorem. $\square$

Concerning the $cAIC^*$ given in (21), let us define $c_1(\theta)$ and $c_2(\theta)$ by

$$c_1(\theta) = \sum_{i=1}^q \text{tr}[(\mathbf{R}\Sigma^{-1})_{(i)}]E[\hat{\theta}_i^{\dagger\dagger}] + \frac{1}{2} \sum_{i=1}^q \sum_{j=1}^q \text{tr}[(\mathbf{R}\Sigma^{-1})_{(ij)}]E[\hat{\theta}_i^\dagger \hat{\theta}_j^\dagger],$$

$$c_2(\theta) = 2 \sum_{i=1}^q \text{tr}[\mathbf{R}\{(\Sigma^{-1})_{(i)} - (\mathbf{R}^{-1})_{(i)}\}]E[\hat{\theta}_i^{\dagger\dagger}] + \sum_{i=1}^q \sum_{j=1}^q \text{tr}[\mathbf{R}\{(\Sigma^{-1})_{(ij)} - (\mathbf{R}^{-1})_{(ij)}\}]E[\hat{\theta}_i^\dagger \hat{\theta}_j^\dagger] \\ + \sum_{i=1}^q \text{tr}[\{2\mathbf{R}(\Sigma^{-1})_{(i)}\Sigma - \mathbf{R}(\mathbf{R}^{-1})_{(i)}\mathbf{R}\}]E[\nabla_{\mathbf{y}} \nabla_{\mathbf{y}}' \hat{\theta}_i^\dagger].$$

Assume the following conditions:

(C3) $\mathbf{R}(\theta)$ is continuously differentiable three times in θ , and $\lim_{N \rightarrow \infty} \lambda_1(\mathbf{R}) > 0$, $\lim_{N \rightarrow \infty} \lambda_N(\mathbf{R}) < \infty$, $\lim_{N \rightarrow \infty} |\lambda_N^i(\mathbf{R})| < \infty$, $\lim_{N \rightarrow \infty} |\lambda_N^{ij}(\mathbf{R})| < \infty$ and $\lim_{N \rightarrow \infty} |\lambda_N^{ijk}(\mathbf{R})| < \infty$, where $\lambda_a(\mathbf{R})$, $\lambda_a^i(\mathbf{R})$, $\lambda_a^{ij}(\mathbf{R})$ and $\lambda_a^{ijk}(\mathbf{R})$ are defined similarly to those for Σ .

(C4) $c_1(\theta)$ and $c_2(\theta)$ are continuously differentiable functions satisfying $\partial_i c_1(\theta) = O(N^{-1})$ and $\partial_i c_2(\theta) = O(N^{-1})$ for $1 \leq i \leq q$.

Theorem A.5. Assume the conditions (A1)–(A4) and (C3)–(C4). Then,

$$E[-2 \log f(\mathbf{y}|\hat{\beta}(\hat{\theta}), \hat{\beta}(\hat{\theta}), \hat{\theta}) - \Delta_{c_1}^*(\mathbf{y}) - \Delta_{c_2}^*(\mathbf{y})] = cAI(\theta) + O(N^{-1/2}),$$

for $\Delta_{c_1}^*(\mathbf{y})$ and $\Delta_{c_2}^*(\mathbf{y})$ given in (20). Also, $E[\Delta_{c_1}^*(\mathbf{y})] = -2\rho(\theta) + O(N^{-1/2})$ and $E[\Delta_{c_2}^*(\mathbf{y})] = c_2(\theta) + O(N^{-1/2})$, where $\rho(\theta)$ is the effective degrees of freedom defined by

$$\rho(\theta) = \text{tr}[(\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}\mathbf{X}'\Sigma^{-1}\mathbf{R}\Sigma^{-1}\mathbf{X}] + N - \text{tr}[\mathbf{R}\Sigma^{-1}].$$

Proof. Define $\Delta_{c1}(\theta)$ and $\Delta_{c2}(\theta)$ by

$$\begin{aligned}\Delta_{c1}(\theta) &= -2 \left\{ E[u' \Sigma^{-1} R \widehat{P}(\widehat{\theta}) u] + N - \text{tr}[R \Sigma^{-1}] \right\}, \\ \Delta_{c2}(\theta) &= E[u' (2 \Sigma^{-1} R \widehat{\Sigma}^{-1} - \Sigma^{-1} R \widehat{R}^{-1} R \Sigma^{-1}) u] - E[\text{tr}[\widehat{R}^{-1} (2R - R \Sigma^{-1} R)]] + 2N - 2\text{tr}[\Sigma^{-1} R].\end{aligned}$$

Using the arguments as in the proof of the Theorem 2.2 in [21], we can show that

$$E_y[-2 \log f(y|\widehat{v}(\widehat{\theta}), \widehat{\beta}(\widehat{\theta}), \widehat{\theta})] - cAI(\theta) = \Delta_{c1}(\theta) + \Delta_{c2}(\theta),$$

and that $\Delta_{c1}(\theta) = -2\rho(\theta) + O(N^{-1/2})$ and $\Delta_{c2}(\theta) = c_2(\theta) + O(N^{-1/2})$, since $\Delta_{c1}(\theta)$ and $\Delta_{c2}(\theta)$ can be expressed as $\Delta_{c1}(\theta) = -2J_3 - 2N + 2\text{tr}[R \Sigma^{-1}]$ and $\Delta_{c2}(\theta) = J_1 - J_2 + 2N - 2\text{tr}[\Sigma^{-1} R]$ based on the notations J_1, J_2 and J_3 in [21]. This implies that $\Delta_{c2}^*(y) = c_2(\widehat{\theta}) + O_p(N^{-1/2})$ for $\Delta_{c2}^*(y)$ given in (20). Since $c_2(\theta) = O(1)$, it is observed that

$$E[\Delta_{c2}^*(y)] = E[c_2(\widehat{\theta})] + O(N^{-1/2}) = c_2(\theta) + O(N^{-1/2}) = \Delta_2(\theta) + O(N^{-1/2}). \quad (\text{A.18})$$

For $\Delta_{c1}(\theta)$, it is noted that $\Delta_{c1}(\theta) = O(N)$ since $\text{tr}[R \Sigma^{-1}] = O(N)$. Kubokawa [20] showed in the proof of his Theorem 2.3 that

$$E[\text{tr}[\widehat{R} \widehat{\Sigma}^{-1}]] = \text{tr}[R \Sigma^{-1}] + c_1(\theta) + O(N^{-1/2}),$$

which leads to $E_*[\text{tr}[\widehat{R}^* \widehat{\Sigma}^{*-1}]|y] = \text{tr}[\widehat{R} \widehat{\Sigma}^{-1}] + c_1(\widehat{\theta}) + O_p(N^{-1/2})$. Then,

$$\begin{aligned}E[2\text{tr}[\widehat{R} \widehat{\Sigma}^{-1}] - E_*[\text{tr}[\widehat{R}^* \widehat{\Sigma}^{*-1}]|y]] &= E[\text{tr}[\widehat{R} \widehat{\Sigma}^{-1}] - c_1(\widehat{\theta})] + O(N^{-1/2}) \\ &= \text{tr}[R \Sigma^{-1}] + O(N^{-1/2}).\end{aligned}$$

Since $E[u' \Sigma^{-1} R \widehat{P}(\widehat{\theta}) u] = \text{tr}[(X' \Sigma^{-1} X)^{-1} X' \Sigma^{-1} R \Sigma^{-1} X] + O(N^{-1/2})$, it is seen that $E_*[u^{*'} \widehat{\Sigma}^{-1} \widehat{R} \widehat{P}(\widehat{\theta}^*) u^*] = \text{tr}[(X' \widehat{\Sigma}^{-1} X)^{-1} X' \widehat{\Sigma}^{-1} \widehat{R} \widehat{\Sigma}^{-1} X] + O_p(N^{-1/2})$. Hence,

$$E[E_*[u^{*'} \widehat{\Sigma}^{-1} R \widehat{P}(\widehat{\theta}^*) u^* + \text{tr}[\widehat{R}^* \widehat{\Sigma}^{*-1}]|y] + N - 2\text{tr}[\widehat{R} \widehat{\Sigma}^{-1}]] = \rho(\theta) + O(N^{-1/2}),$$

that is, $E[\Delta_{c1}^*(y)] = -2\rho(\theta) + O(N^{-1/2}) = \Delta_{c1}(\theta) + O(N^{-1/2})$, which proves the theorem. $\square$

References

- [1] H. Akaike, Information theory and an extension of the maximum likelihood principle, in: B.N. Petrov, F. Csaki (Eds.), 2nd International Symposium on Information Theory, Akademia Kiado, Budapest, 1973, pp. 267–281.
- [2] H. Akaike, A new look at the statistical model identification, system identification and time-series analysis, IEEE Trans. Autom. Contr. AC-19 (1974) 716–723.
- [3] R. Basu, J.K. Ghosh, R. Mukerjee, Empirical Bayes prediction intervals in a normal regression model: higher order asymptotics, Statist. Prob. Lett. 63 (2003) 197–203.
- [4] G.E. Battese, R.M. Harter, W.A. Fuller, An error-components model for prediction of county crop areas using survey and satellite data, J. Amer. Statist. Assoc. 83 (1988) 28–36.
- [5] F.B. Butar, P. Lahiri, On measures of uncertainty of empirical Bayes small-area estimators, J. Stat. Plan. Inference 112 (2003) 63–76.
- [6] S. Chatterjee, P. Lahiri, H. Li, Parametric bootstrap approximation to the distribution of EBLUP and related prediction intervals in linear mixed models, Ann. Statist. 36 (2008) 1221–1245.
- [7] S. Chen, P. Lahiri, On mean squared prediction error estimation in small area estimation problems, Commun. Stat.-Theory Methods 37 (2008) 1792–1798.
- [8] K. Das, J. Jiang, J.N.K. Rao, Mean squared error of empirical predictor, Ann. Statist. 32 (2004) 818–840.
- [9] G.S. Datta, M. Ghosh, D.D. Smith, P. Lahiri, On an asymptotic theory of conditional and unconditional coverage probabilities of empirical Bayes confidence intervals, Scand. J. Stat. 29 (2002) 139–152.
- [10] G.S. Datta, T. Kubokawa, J.N.K. Rao, I. Molina, Estimation of mean squared error of model-based small area estimators, Test 20 (2011) 367–388.
- [11] G.S. Datta, P. Lahiri, A unified measure of uncertainty of estimated best linear unbiased predictors in small area estimation problems, Statist. Sin. 10 (2000) 613–627.
- [12] G.S. Datta, J.N.K. Rao, D.D. Smith, On measuring the variability of small area estimators under a basic area level model, Biometrika 92 (2005) 183–196.
- [13] J.A. Doornik, Object-Oriented Matrix Programming Using Ox, third ed., Timberlake Consultants Press and Oxford, London, 2007.
- [14] J.A. Doornik, M. Ooms, Computational aspects of maximum likelihood estimation of autoregressive fractionally integrated moving average models, Comput. Statist. Data Anal. 41 (2003) 333–348.
- [15] R.E. Fay, R. Herriot, Estimates of income for small places: an application of James–Stein procedures to census data, J. Amer. Statist. Assoc. 74 (1979) 269–277.
- [16] M. Ghosh, J.N.K. Rao, Small area estimation: An appraisal, Statist. Sci. 9 (1994) 55–93.
- [17] P. Hall, T. Maiti, Nonparametric estimation of mean-squared prediction error in nested-error regression models, Ann. Statist. 34 (2006) 1733–1750.
- [18] P. Hall, T. Maiti, On parametric bootstrap methods for small area prediction, J. Roy. Statist. Soc. 68 (2006) 221–238.
- [19] T. Kubokawa, Correction of empirical Bayes confidence intervals in nested error regression models, J. Korean Statist. Soc. 39 (2010) 221–236.
- [20] T. Kubokawa, On measuring uncertainty of small area estimators with higher order accuracy, Discussion Paper Series, CIRJE-F-754, J. Japan Statist. Soc., (2010) (in press).
- [21] T. Kubokawa, Conditional and unconditional methods for selecting variables in linear mixed model, J. Multivariate Anal. 102 (2011) 641–660.
- [22] P. Lahiri, J.N.K. Rao, Robust estimation of mean squared error of small area estimators, J. Amer. Statist. Assoc. 90 (1995) 758–766.
- [23] H. Liang, H.L. Wu, G. Zou, A note on conditional AIC for mixed-effects models, Biometrika 95 (2008) 773–778.
- [24] D. Pfeffermann, Small area estimation—new developments and directions, Int. Statist. Rev. 70 (2002) 125–143.

- [25] N.G.N. Prasad, J.N.K. Rao, The estimation of the mean squared error of small-area estimators, *J. Amer. Statist. Assoc.* 85 (1990) 163–171.
- [26] J.N.K. Rao, *Small Area Estimation*, Wiley, New Jersey, 2003.
- [27] J.N.K. Rao, M. Yu, Small-area estimation by combining time-series and cross-sectional data, *Canad. J. Statist.* 22 (1994) 511–528.
- [28] N. Sugiura, Further analysis of the data by Akaike's information criterion and the finite corrections, *Commun. Statist.—Theory Methods* 1 (1978) 13–26.
- [29] F. Vaida, S. Blanchard, Conditional Akaike information for mixed-effects models, *Biometrika* 92 (2005) 351–370.